

When Is Benchmark Contamination Detectable? Information Limits and Power-Calibrated Audits

Ibne Farabi Shihab^{*†1} and Sanjeda Akter^{*1} and Anuj Sharma²

¹Department of Computer Science, Iowa State University

²Department of Civil, Construction & Environmental Engineering, Iowa State University
ishihab@iastate.edu

Abstract

Behavioral contamination detectors can return “no evidence” either because a benchmark is clean or because the audit has little power. We formalize the distinction for a benchmark in which an unknown fraction α of items was seen during training: with matched clean and seen controls the behavioral channel is the sparse mixture $Q_\alpha = (1 - \alpha)P_0 + \alpha P_1$, and an exact second-moment argument shows detectability is governed by $\alpha\rho\sqrt{m}$, where $\rho^2 = \chi^2(P_1\|P_0)$ measures behavioral separability. Any scalar detector reduces to its efficacy $e_f = |\mathbb{E}_1 f - \mathbb{E}_0 f| / \sqrt{\text{Var}_0(f)} \leq \rho$, which is estimable from controls before the audit is run; a separate sample-split certificate lower-bounds α distribution-free, without an orientation assumption. Our empirical finding is two-sided. Frozen calibration efficacy predicts held-out power curves ($R^2 = 0.83\text{--}0.98$ over six exact-permutation channels), but the efficacy-only Gaussian *budget* is miscalibrated at the small m it prescribes, failing in 9/9 gate-passing channels even though efficacy itself transports—the inversion breaks, not the calibration. A predeclared two-stage planner that simulates the deployed test repairs the budgets, is uniformly conservative, and abstains where its probe does not transport. The certificate is valid but vacuous at audit scale, and a five-seed paired injection study recovers a mechanism ordering (verbatim > paraphrase > surface) in which the apparent answer-only signal is baseline drift. We report the audit contract and its failures together: a non-rejection is interpretable only alongside the efficacy, budget, and validity gates that produced it.

1 Introduction

Public evaluation sets routinely appear in web-scale training corpora. Exact or semantic exposure can inflate a model’s benchmark score without improving the capability the benchmark was

intended to measure (Magar and Schwartz, 2022; Dong et al., 2024). A growing literature therefore tries to infer exposure from token likelihoods, perturbations, output order, or elicited recall (Mattern et al., 2023; Shi et al., 2024; Oren et al., 2024; Golchin and Surdeanu, 2025). Yet empirical studies find that detectors often disagree, fail under distribution shift, or can be evaded (Duan et al., 2024; Fu et al., 2025; Meeus et al., 2025; Samuel et al., 2025; Dekoninck et al., 2024a; Wang et al., 2026a). A 25-model study documents this reliability gap—only 201 of 335 audit outcomes correct, with shift false positives and underpowered post-hoc inference (Zarzecki et al., 2026); it establishes the empirical failure, and our question is whether a calibrated channel can predict it before an audit is run—absent that, a non-rejection is hard to interpret.

The missing object is *audit power*: how many independent items must be probed to detect a small exposed fraction, how does the answer depend on the model’s observable memory of a seen item, and can a detector state, with controlled error, which fractions it rules out? Dataset-level methods aggregate membership scores into significance tests (Maini et al., 2024; Zhang et al., 2024; Puerto et al., 2025), ConStat estimates performance inflation (Dekoninck et al., 2024b), and FTD controls a false discovery rate (Zhang et al., 2026)—important but different deliverables: a valid p -value does not say which exposure fractions the audit had power to detect, and inflation is not the fraction exposed.

We study the benchmark-level hypothesis $H_0 : \alpha = 0$ versus $H_1 : \alpha > 0$, where α is the fraction of benchmark items exposed during training. A probe outcome Y can contain a scalar loss, a vector of token statistics, a set of stochastic generations, or a black-box answer. Conditional on an item being clean or seen, its outcome follows P_0 or P_1 . Uniformly sampling benchmark items gives the

^{*}Equal contribution.

[†]Corresponding author: ishihab@iastate.edu.

mixture

$$Q_\alpha = (1 - \alpha)P_0 + \alpha P_1. \quad (1)$$

This assumption is substantive—it requires matched clean controls and rules out unmodeled spillover—and we state it visibly, test its implications, and show what fails without it.

Contributions.

1. We connect behavioral contamination audits to classical sparse-mixture detection (Ingster, 1997; Donoho and Jin, 2004; Cai et al., 2011; Cai and Wu, 2014), instantiating a sharp second-moment limit in the observable $\rho^2 = \chi^2(P_1 \| P_0)$ for Bernoulli exposure and fixed contaminated subsets; we claim the audit interpretation, not a new mixture boundary.
2. We operationalize that limit for arbitrary contamination scores: the efficacy $e_f \leq \rho$ of the oriented mean-score audit built from f yields a prospective *local-asymptotic* power prediction and a finite-sample lower bound on channel separability. Power *curves* transport across splits (R^2 up to 0.99); the efficacy-only Gaussian *planner* is miscalibrated at the small budgets it prescribes, and we report that failure with its diagnosis rather than a pooled scaling law (§6.2).
3. We provide a sample-split audit reporting a distribution-free lower confidence bound on the exposed fraction α , with coverage for any independently learned bounded score, even one accidentally reversed.
4. We design complementary evaluations: checkpoint-matched channels, gate-passing same-corpus continued-pretraining channels with direct budget trials, and causal injections with five paired clean-continuation seeds. Primary falsifiable endpoints: prospective power prediction, efficacy transport (tested; efficacy itself largely transports, but the Gaussian efficacy-to-budget *inversion* fails at prescribed budgets), and coverage.

2 Problem Formulation

2.1 From benchmark items to a behavioral channel

Fix a trained language model, a benchmark population, an access regime, and a predeclared probing procedure. Let $Z \in \{0, 1\}$ indicate whether a uniformly sampled benchmark item was exposed

during training. The complete outcome from that item is $Y \in \mathcal{Y}$. We treat all repeated decodes, perturbations, and token-level features from the *same* item as one joint outcome; counting them as independent probes would artificially inflate m .

Assumption 1 (Matched partial-contamination channel). For the audited model and probe, $Y \mid Z = 0 \sim P_0$ and $Y \mid Z = 1 \sim P_1$. Probe items are sampled independently and uniformly, $Z \sim \text{Bernoulli}(\alpha)$, and the conditional laws P_0, P_1 do not change with α over the range being audited.

The last clause is a local no-spillover/stability condition: it does not claim training is literally itemwise, but specifies when a clean reference channel transports to the suspect model. Clean-control and held-out items measure violations in section 5; heterogeneous and adaptive versions are in section E.

Under theorem 1, m outcomes follow $P_0^{\otimes m}$ under H_0 and $Q_\alpha^{\otimes m}$ under H_1 . A possibly randomized audit is a measurable $\phi_m : \mathcal{Y}^m \rightarrow [0, 1]$, where $\phi_m = 1$ means “detect contamination,” with type-I error $a_m(\phi_m) = \mathbb{E}_0[\phi_m]$, type-II error $b_m(\phi_m, \alpha) = \mathbb{E}_\alpha[1 - \phi_m]$, and power $\pi_m(\phi_m, \alpha) = 1 - b_m(\phi_m, \alpha)$. We call an audit level τ if $a_m \leq \tau$.

2.2 Why contamination fraction alone is insufficient

There is no nontrivial guarantee that depends only on (α, m) : setting $P_1 = P_0$ makes the clean and contaminated transcript laws identical ($a_m + b_m = 1$ for every test), representing exposure that leaves no trace in the chosen access channel, and an unconstrained clean reference law lets any observed law be relabeled as clean. A universal $\Theta(m^{-1/2})$ threshold therefore needs an explicit informativeness parameter.

Definition 2 (Behavioral separability). Assume $P_1 \ll P_0$, let $r = dP_1 / dP_0$, and define

$$\begin{aligned} \rho^2 &:= \chi^2(P_1 \| P_0) \\ &= \mathbb{E}_0[(r(Y) - 1)^2]. \end{aligned} \quad (2)$$

ρ depends jointly on the model, mechanism, benchmark distribution, probe, and access regime: zero when the probe carries no membership signal, large when seen items behave in ways rare under P_0 , and infinite off-support (where a single witness beats the regular rate). Our matching result concerns the practically common overlapping, finite- ρ regime.

3 Fundamental Detection Limit

3.1 A finite-sample second-moment bound

For compactness, define the second-moment radius

$$C_m(\alpha, \rho) := \frac{1}{2} \{(1 + \alpha^2 \rho^2)^m - 1\}^{1/2}. \quad (3)$$

Theorem 3 (Power limit for partial contamination). *Under theorem 1, suppose $P_1 \ll P_0$ and $\rho^2 < \infty$. For every $\alpha \in [0, 1]$, every $m \geq 1$, and every possibly randomized test ϕ_m ,*

$$a_m(\phi_m) + b_m(\phi_m, \alpha) \geq 1 - \min\{1, C_m(\alpha, \rho)\}. \quad (4)$$

Consequently, every level- τ audit satisfies

$$\pi_m(\phi_m, \alpha) \leq \min\{1, \tau + \min\{1, C_m(\alpha, \rho)\}\}. \quad (5)$$

The calculation is exact up to the final TV inequality ($\chi^2(Q_\alpha^{\otimes m} \| P_0^{\otimes m}) = (1 + \alpha^2 \rho^2)^m - 1$, then $\text{TV} \leq \frac{1}{2} \sqrt{\chi^2}$), sharper than passing through KL and Pinsker; all steps are in section A.1.

The same limit covers the realistic design in which an auditor evaluates every item of a benchmark with a *fixed, unknown* contaminated subset of size k : theorem 6 (stated and proved in section A.2) gives the analogous bound with weak-signal boundary $k\rho/\sqrt{m} = \alpha\rho\sqrt{m} = \Theta(1)$ for $k = \alpha m$, the symmetric oracle score attains it without knowing the subset, and the subset-averaged bound is a Bayes and hence minimax lower bound.

Corollary 4 (Regular detection boundary). *For any sequence of channels and alternatives for which $\alpha_m \rho_m \sqrt{m} \rightarrow 0$, every asymptotic level- τ audit has $\limsup_m \pi_m(\alpha_m) \leq \tau$. Thus it has no asymptotic power beyond its false-positive rate. A necessary order for nontrivial power is*

$$\alpha_m = \Omega\left(\frac{1}{\rho_m \sqrt{m}}\right).$$

The familiar $m^{-1/2}$ exponent appears only when ρ_m remains bounded away from zero and infinity; capacity alone does not upper-bound ρ without a training-stability theorem, so the informal capacity step of prior arguments is replaced by an observable channel quantity.

3.2 A matching test

The lower-bound scaling is attainable: the centered likelihood-ratio score $g = r - 1$ yields a

test that is locally asymptotically optimal, attaining the envelope $1 - \Phi(z_{1-\tau} - h)$ at local alternatives $\alpha_m = h/(\rho\sqrt{m})$, with the oracle sample budget $m_{\text{oracle}} \approx ((z_{1-\tau} + z_{1-\beta})/(\alpha\rho))^2$ (eq. (18)); every regular channel admits a test matching the $\alpha\rho\sqrt{m}$ boundary (construction and LAN statement in section D).

3.3 What an existing detector score can achieve

Most practical methods output a scalar score rather than an estimated density ratio. The following proposition turns such a score into a power diagnostic.

Proposition 5 (Score efficacy). *For any nonconstant $f \in L^2(P_0)$, let*

$$\Delta_f = \mathbb{E}_1 f - \mathbb{E}_0 f, \quad \sigma_f^2 = \text{Var}_0(f),$$

$$e_f = \frac{|\Delta_f|}{\sigma_f}.$$

Then $e_f \leq \rho$, with equality if and only if $f - \mathbb{E}_0 f = c(r - 1) P_0$ -almost surely for some nonzero constant c . Moreover,

$$\mathbb{E}_{Q_\alpha} f - \mathbb{E}_0 f = \alpha \Delta_f,$$

so the standardized local shift of the mean-score audit is $\alpha\sqrt{m} e_f$.

Thus AUROC and benchmark-level power answer different questions: two scores with similar AUROC can have different low- α efficacy, weak instance-level separation can still accumulate over a large benchmark, and $e_f/\rho \in [0, 1]$ measures how much local information a score uses. For an oriented mean-score test, the operational planning equation is

$$m_{\text{target}}(f) \approx \left(\frac{z_{1-\tau} + z_{1-\beta}}{\alpha e_f}\right)^2. \quad (6)$$

Unlike eq. (18), this quantity is estimable from matched controls and refers to the chosen detector, not an unattained channel oracle. Two scope warnings, borne out empirically: e_f summarizes the local power of the *oriented mean-score test* built from f , not of every detector computable from f ; and eq. (6) is a first-order Gaussian inversion, inheriting every finite-sample defect of the local approximation (discreteness, higher moments, alternative variance, test conservatism)—§6.2 measures exactly this failure at small m .

A companion variance-adaptive finite-sample floor (theorem 9, section C) converts calibration samples of a bounded score into a certified lower bound $\rho \geq e_f \geq \underline{e}_f$ that holds with probability $1 - \delta_{\text{cal}}$, using the empirical-Bernstein standard-deviation radius of Maurer and Pontil (2009). Substituting $\underline{e}_f$ into eq. (6) is conservative about calibration uncertainty but remains a local-asymptotic planning calculation, not a finite-sample power guarantee.

3.4 Adaptive and heterogeneous probes

An auditor may choose the next prompt after observing previous answers. Adaptivity can allocate queries toward more informative probes but cannot exceed a cumulative KL information budget of $\frac{1}{2} \sum_t \log(1 + \alpha^2 \rho_t^2)$ over fresh items (theorem 10 and section E); independent heterogeneous probes admit exact chi-square tensorization, with local information $\alpha^2 \sum_i \rho_i^2$. The precise statement and the KL/Pinsker power bound appear in section E.

4 A Power-Calibrated Audit

4.1 Learning an efficient score

The oracle $r = p_1/p_0$ is unknown, so we learn a probabilistic classifier $s_\theta(y) \approx \mathbb{P}(Z = 1 \mid Y = y)$ from labeled, matched clean and seen controls under balanced sampling; the primary audit uses the clipped score $f_\theta(y) \in [0, 1]$ for stability and finite-sample certification, with the unclipped implied ratio $s_\theta/(1 - s_\theta)$ kept only as a cross-fitted diagnostic, never as an oracle estimate of ρ .

To prevent optimistic power estimates, data are divided by original document or benchmark item into a *score-training split* (fits f_θ), a *calibration split* (estimates $\mathbb{E}_0 f$, $\mathbb{E}_1 f$, σ_f , and prospective budgets), and an untouched *audit split* with fresh control items for testing and confidence bounds. Cross-fitting can rotate these roles, but every reported fold keeps the score independent of its calibration and audit observations. Existing scores (loss, Min-K, ReCaLL, and others) enter exactly the same pipeline after their orientation is chosen on the training split.

4.2 Detection and prospective power

Let $\hat{\Delta}_f = \bar{f}_1 - \bar{f}_0$ and $\hat{\sigma}_f$ be computed only on calibration controls. The plug-in efficacy $\hat{e}_f = |\hat{\Delta}_f|/\hat{\sigma}_f$ gives the point budget in eq. (6); replacing it by $\underline{e}_f$ gives a one-sided, uncertainty-aware local budget. The *primary confirmatory*

decision is a one-sided two-sample permutation test of $\hat{s}_f(\bar{f}_A - \bar{f}_0^{\text{test}})$, where $\hat{s}_f = \text{sign}(\hat{\Delta}_f)$ and all score choices and orientations are frozen before the audit split is opened. Conditional on exchangeability of the pooled audit/clean scores under H_0 and on the frozen score, the exact (randomized) permutation test controls type-I error at $\tau = 0.05$. The deployed implementation is Monte-Carlo with 199 permutations and the add-one estimator $\hat{p} = (1 + \#\{\text{perm} \geq \text{obs}\})/200$ (valid; conservative on a $1/200$ grid); exchangeability is exactly what the validity gates (§4.4) probe, so empirical size is reported per channel, never asserted (the initial runs use a cheaper standardized-mean approximation, making size a measured endpoint). Our remaining empirical endpoints are power as a function of (α, m) and its prediction from the frozen calibration efficacy (R^2 , MAE; table 7); predicted versus observed local sample-size targets for 80% power (table 10); and the smallest detectable fraction at the available size (table 11). Bootstrap intervals resample original items, never individual tokens or decodes.

When the null mean is estimated from n_0^{test} fresh clean controls, the local coordinate uses $m_{\text{eff}} = (m^{-1} + (n_0^{\text{test}})^{-1})^{-1}$ in place of m , and every main power curve accounts for it; equations below write m for the externally calibrated or $n_0^{\text{test}} \gg m$ regime.

At $\tau = .05$ and 80% power, $z_{.95} + z_{.80} = 2.486$. Thus

$$\alpha_{\min}(f, m) \approx \frac{2.486}{e_f \sqrt{m}}. \quad (7)$$

Table 11 makes the resulting sensitivity analysis usable when a closed model does not permit matched control construction.

4.3 A finite-sample lower confidence bound

A test can say that contamination is present; a prevalence certificate says how much, distribution-free for a bounded, independently learned score.

Theorem 8 (stated formally in section B, proved in section A.7) takes any $[0, 1]$ -valued score f fixed independently of the data, three mutually independent samples— m audit items, n_0 clean controls, n_1 seen controls—and returns

$$\underline{\alpha} = \begin{cases} \min\{1, \max\{0, \frac{N_L}{D_U}\}\}, & D_U > 0, \\ 0, & D_U \leq 0, \end{cases} \quad (8)$$

with $\varepsilon_{\text{cert}}(n) = \sqrt{\log(6/\delta_{\text{cert}})/(2n)}$, where $N_L = (\bar{f}_A - \varepsilon_{\text{cert}}(m)) - (\bar{f}_0 + \varepsilon_{\text{cert}}(n_0))$ shrinks

the observed audit-clean gap by *both* sample radii and $D_U = (\bar{f}_1 + \varepsilon_{\text{cert}}(n_1)) - (\bar{f}_0 - \varepsilon_{\text{cert}}(n_0))$ inflates the control gap; then $\mathbb{P}_\alpha\{\underline{\alpha} \leq \alpha\} \geq 1 - \delta_{\text{cert}}$ by a simultaneous three-sample Hoeffding argument. The $D_U \leq 0$ guard matches the formal statement in section B: without it, a reversed score can give $N_L < 0$ and $D_U < 0$ with a spuriously positive ratio. No orientation assumption is needed (a reversed score yields $\underline{\alpha} = 0$), the bound is conservative by design, and rejecting when $\underline{\alpha} > 0$ is a fallback whose rejection rate we report beside the confirmatory permutation test, not the source of the main power curves.

4.4 Required validity checks

The mathematics cannot rescue a mismatched control channel. We therefore make four diagnostics part of the audit contract: **blind separation** (a text-only classifier must not separate the pools), **clean-channel transport** (clean-control scores must not shift across runs), **channel stability** (conditional score laws must be stable across α), and the **independence unit** (effective m counts source-item clusters, never near-duplicates or repeated decodes). Full definitions and thresholds are in section F.

The interpretation is also predeclared: a pooled transport claim requires median multiplicative sample-size error ≤ 2 and $\geq 80\%$ of held-out channels inside nominal 90% prediction intervals; otherwise the pooled scaling claim is withdrawn and calibration-to-audit nontransport becomes the primary conclusion. (The v5 intervals are descriptive, so the second criterion is unevaluable as specified; §6.2.)

5 Experiments

5.1 Research questions

Four pre-specified questions: **RQ1 (prospective prediction)**—does calibration-set efficacy predict held-out audit power and sample size? **RQ2 (certification)**—does eq. (8) attain nominal coverage, and how conservative is it? **RQ3 (transport and mechanism)**—how do mechanism, repetition, access, and post-training alter $\hat{e}_f$, its floor, and rankings? **RQ4 (scaling diagnostic)**—does power collapse on $\alpha \hat{e}_f \sqrt{m_{\text{eff}}}$ with slope near $-1/2$?

5.2 Evaluation A: checkpoint-matched channels

The design follows the checkpoint construction of Wang et al. (2026b): members from a win-

dow before checkpoint step t , nonmembers from a matched unprocessed window, balanced on source, length, packing, deduplication, and stream position, blind text-only baseline first. The completed runs use Pythia (Biderman et al., 2023) and GPT-Neo; in the six-channel audit the member and nonmember pools come from *different corpora*, so all six fail the blind gate (§6.4)—the gate working as designed. OLMo (Groeneveld et al., 2024) channels are planned, not run.

For each held-out channel, we create aggregate audit sets by sampling a known fraction α of members and $1 - \alpha$ nonmembers. This tests the statistical audit conditional on a real model channel; it does *not* claim the post-hoc mixture retrains the model or validates no-spillover—that causal question is Evaluation B’s.

Gate-passing channels (v3). A third audit randomly splits one corpus (700+700) and continues pretraining each base model on the member half: pools text-exchangeable by construction, membership causally induced, with *direct* power trials at the frozen budgets (all details in table 3).

Locked external validation. A locked validation on the open 25-model release of Zarzecki et al. (2026)—efficacy calibrated without their audit outcomes predicting their documented failures, budgets, and coverage, no score selection or recalibration—is a planned extension, not a completed result; the Zarzecki transport is not yet run.

5.3 Evaluation B: controlled contamination injection

We distinguish two contamination fractions that are easy to conflate: the *training injection fraction* β_{train} (items injected during continuation over benchmark size), a causal intervention whose variation requires independently trained models, and the *audit exposed fraction* α_{audit} (exposed items over audit-set size), which can be varied by post-hoc remixing conditional on one trained model. Remixing α_{audit} measures audit power, not the causal effect of more training contamination; they coincide only when the entire benchmark is audited.

Starting from the clean EleutherAI/pythia-160m checkpoint (Pile-trained; see the provenance caveat below), we continue training for a fixed token and optimizer budget (400 AdamW steps, learning rate 5×10^{-5}). We inject a β_{train} fraction of SQuAD benchmark items into a Pile background corpus and audit against a disjoint held-out pool of the

same benchmark; the main injected condition uses 200 injected and 200 held-out items. Total updates, token count, and order randomization are matched to a clean ($\beta_{\text{train}} = 0$) run per training seed.

We cross the injection with four mechanisms at $k = 4$ copies per injected item: exact question–answer text; a *validated paraphrase* (answer preserved, overlap-capped, dual independent NLI judges; acceptance $138/200 = 0.69$; pipeline in section G.2); a word-order shuffle (a *surface perturbation*); and answer/rationale-only exposure. Auditing both exposed and held-out items makes spillover directly observable. The upgraded design runs *five* paired training seeds, each with a matched clean ($\beta_{\text{train}}=0$) continuation (same background, order, steps, optimizer, initialization); the causal readout is the per-seed *paired contrast* $D_s = (\text{exposed} - \text{held gap, contaminated}) - (\text{same gap, clean})$ in clean-model units, with item-level bootstrap intervals. Larger exposure counts remain future work. “Pile-trained” does not certify the base checkpoint never saw SQuAD, so we claim only the *incremental* exposure effect of the paired continuations—a well-defined causal quantity even with prior exposure (full caveat in section G.3).

5.4 Probes and statistical protocol

All reported experiments evaluate five gray-box probe families (mean loss, zlib-normalized loss, Min-K% Prob, Min-K%++, neighborhood loss); every “best probe” is *the probe with the highest calibration-split efficacy*, fixed before audit power is computed. ReCaLL, CAMIA-style, black-box, and cross-fitted-combination probes are *specified but not run* (section G); no result depends on them. The nominal confirmatory size is $\tau = 0.05$, the certificate miscoverage is $\delta_{\text{cert}} = 0.05$, and the calibration-floor failure probability is $\delta_{\text{cal}} = 0.05$; these are distinct guarantees. The target power is 0.80. Probe counts are $m \in \{16, \dots, 1024\}$ truncated at the number of independent item clusters, with a fresh clean reference of $n_0^{\text{test}} = 250$ items and m_{eff} reported per test. Monte-Carlo replication counts, interval conventions, onset-fitting rules, and the Holm-adjusted secondary analyses are specified in section G; the transport decision uses the predeclared criteria of section 4.4.

6 Results

6.1 Calibration predicts held-out power

The initial audit freezes $\hat{e}_f$ on the calibration split of one 500+500 Pythia-1.4b channel ($R^2 = 0.81$ – 0.99 , MAE 0.02–0.10; tables 7 and 10; no fitted recalibration on the audit split). The primary evidence is an upgraded audit using the *exact* Monte-Carlo permutation test ($n_0=250$ reference, audits drawn without replacement) over six model×corpus channels (pythia-410m/1.4b, GPT-Neo-1.3B; news and Wikipedia pools of 1,600+1,600). Figure 1 shows every predicted-vs-observed cell with Wilson intervals. Per channel (best probe): $R^2 = 0.83$ – 0.98 , pooled MAE 0.03–0.11, slope 0.91–1.04, intercept -0.09 – 0.06 , 0.8-boundary accuracy 0.95–1.00; in the informative 0.2–0.9 band MAE grows to 0.06–0.25—mid-band prediction is real but coarser. Empirical size at $\alpha=0$ is 0.009–0.045 (table 2): valid, somewhat conservative. Scope, per our own protocol: these six channels fail the blind gate (§6.4), so their R^2 validates power-curve prediction *within a content-confounded channel*; the confirmatory reading rests on the gate-passing v3 channels, and accurate curves do *not* license plug-in budgets (§6.2); values in the fig. 1 caption.

6.2 Plug-in budgets fail: the efficacy-only planner is miscalibrated at small m

Accurate power *curves* do not imply usable plug-in *budgets*. The pre-specified criterion (§4.4) fails on the initial channel: table 10 reports *efficacy-transport ratios* $\exp(2|\log \hat{e}_f^{\text{cal}} - \log \hat{e}_f^{\text{test}}|)$ with median $2.23\times$ and 3/5 channels above the $2\times$ tolerance, so **we withdraw the pooled calibration-to-audit scaling claim** (tolerance fixed before the numbers were seen). But with 800-item calibration halves the ratios shrink to 1.03–1.50, and in the v3 gate-passing channels the *raw* efficacy ratios are 1.02–1.35 (1.04–1.81 squared to the budget scale $m \propto e_f^{-2}$)—within the $2\times$ tolerance. Efficacy itself largely transports; that is *not* what breaks the budgets.

The decisive diagnosis comes from the v3 censoring-free direct trials (table 3; the earlier dyadic-grid “0/3” claim is withdrawn as interval-censored, arithmetic in table 2). At the frozen m_{pred} the planner predicts 0.80 by construction; observed power is 0.67, 0.56, 0.62, Wilson intervals excluding 0.80 in all three channels; re-running the *same* plug-in with the audit-split effi-

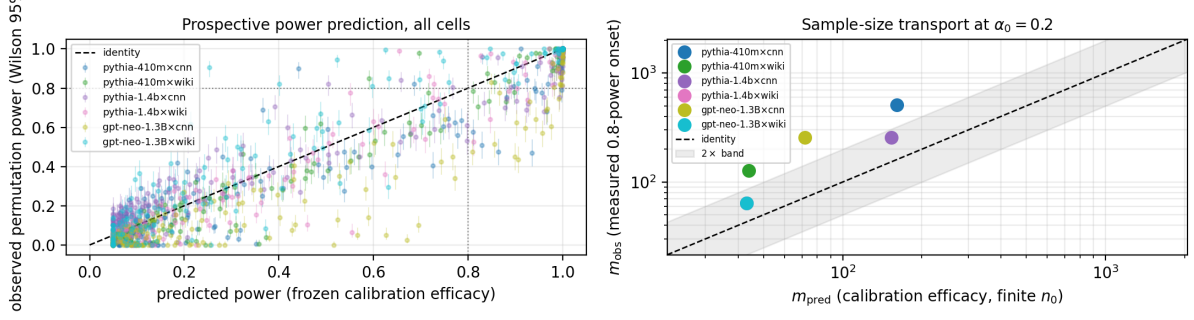


Figure 1: Upgraded six-channel audit, exact permutation test. Left: predicted vs. observed power, every (channel, probe, α , m) cell, Wilson intervals. Right: m_{pred} vs. measured m_{obs} at $\alpha_0=0.2$ (log-log, $2 \times$ band); the dyadic-grid onsets are *interval-censored upper bounds*, so the apparent $1.5\text{--}3.6 \times$ ratios are not resolved onset ratios (two of three cases are unresolved; §6.2)—the censoring-free diagnosis comes from the v3 direct trials.

cacy still predicts 0.58, 0.77, 0.68—errors of both signs up to 0.21 that dwarf the efficacy shift (on pythia-410m $\times$ wiki the efficacies nearly coincide, 4.62/4.31, yet the plug-in predicts 0.77 against 0.56). The failure is **not** calibration-to-audit transport; it is *consistent with* the local Gaussian approximation breaking at the small budgets it prescribes ($m = 8\text{--}42$) plus a conservative exact test (size 0.000–0.013), but the causal decomposition is not isolated (no skewness/tie or Gaussian-vs-Welch-vs-permutation ablation was run; open). At $m_{\text{infl}} = 1.40 \times m_{\text{pred}}$ (frozen before scoring; uncertainty not propagated, disclosed): two clear failures, one unresolved (0.76 [0.70, 0.81]).

The constructive repair is *executed at scale*. A **finite-sample planner** resamples full P_0/P_1 score laws from calibration data only and runs the deployed permutation test at each candidate m . The v4 pilot (table 4) beat the Gaussian budget head-to-head but chose m from pointwise bootstrap bounds and retained only two channels. The v5 protocol repairs both defects (table 5): 12 same-corpus channels, an equivalence blind gate (9/12 pass the predeclared one-sided form, AUC upper CI ≤ 0.55 ; 8/12 pass the stricter two-sided sensitivity gate, which additionally excludes exactly the channel where the planner abstains; per-channel values in table 6), a two-stage planner (stage-A bootstrap candidate m_A on one calibration half; stage-B confirmation on the untouched half over the pre-specified ladder $\{m_A, \lceil 1.5m_A \rceil, 2m_A\}$ with Bonferroni-corrected one-sided bounds), and the **prespecified transport endpoint** on the audit half (estimated onset $\hat{m}_\star$, multiplicative errors, 90% bootstrap quantile intervals). *Validity scope, stated exactly*: the Bonferroni step makes stage B valid against *rung selection* conditional on the empirical

cal-B distribution (175 clean clusters resampled with replacement to $n_0=250$), excluding score-law uncertainty, cluster reuse, and cal-to-audit transport (measured by the audit endpoint, not certified); $\hat{m}_\star$ is a Monte-Carlo onset estimate (200 trials), the 90% intervals are descriptive ($B=30$, probe selection not re-run), the ratios and 8/8 coverage inherit that noise ($\hat{\pi}(m_{\text{plan}}) = 0.785$ at $m_{\text{plan}} = \hat{m}_\star = 56$ on pythia-70m $\times$ cnn), and the prespecified $\geq 80\%$ interval-coverage criterion is *unevaluable as specified* and not claimed. Results: the Gaussian budget *clearly fails in 9/9* channels (power 0.01–0.74 at m_{pred} , every Wilson interval excluding 0.80; $m_{\text{pred}}/\hat{m}_\star = 0.54\text{--}0.86$, median 0.59); the planner is uniformly conservative ($m_{\text{plan}}/\hat{m}_\star = 1.00\text{--}2.00$, median 1.36), delivering 0.785–0.92 power (≥ 0.80 in 6/8, two unresolved, zero clear failures; 90% intervals cover $\hat{m}_\star$ in 8/8). On pythia-70m $\times$ wiki the planner *abstains* (stage A never certifies 0.80) while the Gaussian budget prescribes $m=22$ at 0.01 power—abstention is the designed behavior, and the two-sided gate excludes this channel outright. The certificate (below) makes no planning assumption and the injection ordering (§6.4) stands.

6.3 Certificate coverage: valid, and vacuous at audit scale

The scaled-slack heuristic’s 99.0–99.5% endpoints (table 9) are not validation of the distribution-free certificate, which never fires at $m=512$ (zero false, zero nonzero in all six channels; coverage vacuously 100%): certified $\alpha=0.1\text{--}0.2$ needs balanced groups of $\sim 3,500\text{--}79,000$ items at measured margins. A nonzero $\underline{\alpha}$ is trustworthy; a zero is uninformative—at audit sizes the certificate is a null instrument, reported as a limitation (full analy-

sis: sections H to J).

6.4 Mechanism, access, and validity diagnostics

The initial three-seed injections already show the mechanism decay (0.741/0.343/0.150 exact/surface/answer-only, 3/3 seeds; table 9).

The five-seed *paired-contrast* rerun strengthens this, using *one common frozen probe* (mean NLL, the modal calibration winner) since the per-cell best probe varies by seed and mechanism. The contrasts D_s (item-bootstrap 95% intervals excluding zero in all 15 seed $\times$ mechanism cells) are 1.01 ± 0.07 exact, 0.70 ± 0.09 paraphrase, 0.44 ± 0.04 surface, ordering intact in 5/5 seeds; answer-only has $D_s = -0.01 \pm 0.07$ with every per-seed interval covering zero—its apparent raw efficacy (0.19) is matched by the clean continuation (baseline drift, the artifact the paired design exists to catch), and it is where exchangeability degrades (pooled size 0.087 vs. 0.013–0.058). We claim only “indistinguishable from clean continuation at $n=200$, $k=4$ ” for answer-only; the exact certificate at injection scale never fires (zero false, zero nonzero, coverage 1.0), matching section J.

Every per-seed D_s , interval, size-gate value, and adjacent paired difference is in table 1; on the common probe all three adjacent differences (0.31 ± 0.12 , 0.26 ± 0.10 , 0.45 ± 0.08) are positive in 5/5 seeds (sign-test $p = 0.031$ each; Holm $p_{\text{adj}} = 0.094$: a consistent descriptive pattern, not individually significant after correction, and—as the arms are not dose-matched—not a causal mechanism comparison). The exposure-effect claim is scoped to pythia-160m, SQuAD, $k=4$, this protocol.

The exact-injection channel also compares plug-in predictions against observed 0.8-power onsets: per seed m_{pred} of 371, 208, 301 vs. m_{obs} of 512, 256, 512 (dyadic grid, quantized upward), within the $2\times$ tolerance in all three seeds; surface and answer-only never reach 0.8 power on the tested grid—censored, not extrapolated. Efficacies are estimates or lower bounds on ρ ; the small-sample floor vacuity at $n=200$ is in table 13.

The validity gates are measured per channel (table 2): every checkpoint-free Evaluation A channel *fails* blind separation (text-only AUC 0.95—content-confounded separability, exactly what the gate exposes), while the v3 same-corpus channels repair it (blind AUC 0.48–0.54; 4/6 pass all gates, the two Pythia $\times$ SQuAD channels failing the size

gate, table 3). The v3 rule is not an equivalence test; requiring upper CI ≤ 0.55 also excludes pythia-410m $\times$ wiki (upper 0.579), leaving the small- m conclusion on the two surviving channels (0.67, 0.62, both excluding 0.80). Pure-memorization claims rest on the gate-passing v3 channels and the Evaluation B injections.

7 Related Work

Behavioral detectors infer exposure from likelihoods, perturbations, output order, or elicited recall (Mattern et al., 2023; Shi et al., 2024; Oren et al., 2024, *inter alia*), and evaluations of them report a reliability gap rather than a ranking (Duan et al., 2024; Das et al., 2025; Zarzecki et al., 2026): they establish *that* audits fail, not a quantity—measurable before an audit—saying when they will. Dataset-level methods aggregate item scores into p -values, inflation estimates, or FDR-controlled decisions (Maini et al., 2024; Dekoninck et al., 2024b; Zhang et al., 2026); none states which α the audit had power to detect. The $\alpha\rho\sqrt{m}$ boundary and prevalence estimation are classical (Ingster, 1997; Cai and Wu, 2014; Scott, 2015); we claim audit deliverables, not new mixture theory. Appendix N gives the full treatment, including a per-method comparison of estimands (Table 15).

8 Discussion and Conclusion

Frozen calibration efficacy predicts held-out audit power ($R^2 = 0.83$ – 0.98), but its Gaussian inversion into a budget does not: the plug-in fails in 9/9 gate-passing channels ($m_{\text{pred}}/\hat{m}_* = 0.54$ – 0.86) even though efficacy itself transports (ratios 1.02–1.35), so the culprit is the local approximation at the small m it prescribes, not calibration drift. Simulating the deployed test instead yields budgets that are uniformly conservative, deliver 0.785–0.92 power, and abstain where the probe fails to transport. A non-rejection is thus interpretable only with its efficacy, budget, and gate values attached—and the prevalence certificate, though valid, is vacuous at audit scale.

Limitations

The mixture channel is an auditable modeling assumption, not a universal law of training. Contaminated examples can affect unexposed examples, and post-training can alter the clean and seen channels. Our stability diagnostics can reveal large violations but cannot prove that a clean reference is

perfectly transportable. The lower bound is point-wise in a chosen behavioral access channel; it does not apply to direct corpus search, cryptographic provenance, or support-separated watermark evidence. The matching optimality result is local and assumes finite chi-square divergence and a mild moment condition. The distribution-free prevalence certificate requires independent matched controls and a score learned on separate data; using the audit set to tune the score invalidates its coverage. Such controls are generally unavailable to an external auditor of a closed model, so neither efficacy nor the exposed fraction is then identified without extra assumptions. Finally, contamination encompasses more than verbatim membership. Semantic task exposure, answer-only exposure, and distillation may require different labels and channels, so results should not be transported between mechanisms without validation. Six specific gaps remain open and unexecuted: the v5 planner’s stage-B guarantee is conditional on the empirical cal-B distribution (a population-level selection-valid planner would need to propagate score-law uncertainty and cluster reuse, e.g. by nesting the bootstrap over probe selection with a larger replicate budget); the prespecified interval-coverage transport criterion is unevaluable with descriptive intervals and is not claimed; the as-run size gate has the detection rather than equivalence direction (the equivalence form leaves 4/12 channels, table 6); the cause of Gaussian miscalibration is consistent with small- m non-Gaussianity plus test conservatism but not isolated by ablation; the mechanism ordering is not dose-matched (a causal comparison requires retraining all arms on the 138-item paraphrase intersection with copy- and token-matched exposure); and no locked external or modern-model validation (OLMo-class models, black-box access, the Zarzecki study, or matched-counterpart designs such as LLM Dataset Inference or PaCoST) has been run. The certificate remains a theoretical contribution: mathematically valid but firing on no reported condition. Where these gaps would change a claim, the claim is scoped accordingly.

Ethical Considerations

Contamination audits can improve scientific accountability, but membership signals can also expose whether copyrighted, private, or sensitive text was used for training. Experiments should use licensed or public data, avoid releasing recon-

structable private examples, and report aggregate statistics. The proposed lower bound must not be used to imply that undetected contamination is acceptable. Controlled contamination runs should be confined to research models and should not publish contaminated checkpoints as general-purpose models without prominent documentation. All experimental results, compute use, data licenses, and automated writing or coding assistance must be disclosed in the venue’s Responsible NLP checklist.

References

- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar Van Der Wal. 2023. Pythia: A suite for analyzing large language models across training and scaling. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*.
- Gilles Blanchard, Gyemin Lee, and Clayton Scott. 2010. Semi-supervised novelty detection. *Journal of Machine Learning Research*, 11:2973–3009.
- T. Tony Cai, X. Jessie Jeng, and Jiashun Jin. 2011. Optimal detection of heterogeneous and heteroscedastic mixtures. *Journal of the Royal Statistical Society: Series B*, 73(5):629–662.
- T. Tony Cai and Yihong Wu. 2014. Optimal detection of sparse mixtures against a given null distribution. *IEEE Transactions on Information Theory*, 60(4):2217–2232.
- Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Ulfar Erlingsson, Alina Oprea, and Colin Raffel. 2021. Extracting training data from large language models. In *30th USENIX Security Symposium*, pages 2633–2650.
- Hongyan Chang, Ali Shahin Shamsabadi, Kleomenis Katevas, Hamed Haddadi, and Reza Shokri. 2025. Context-aware membership inference attacks against pre-trained language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Debeshee Das, Jie Zhang, and Florian Tramèr. 2025. Blind baselines beat membership inference attacks for foundation models. *arXiv preprint arXiv:2406.16201*.
- Jasper Dekoninck, Mark Niklas Müller, Maximilian Baader, Marc Fischer, and Martin Vechev. 2024a. Evading data contamination detection for language models is (too) easy. *arXiv preprint arXiv:2402.02823*.

- Jasper Dekoninck, Mark Niklas Müller, and Martin Vechev. 2024b. ConStat: Performance-based contamination detection in large language models. In *Advances in Neural Information Processing Systems*, volume 37.
- Yihong Dong, Xue Jiang, Huanyu Liu, Zhi Jin, Bin Gu, Mengfei Yang, and Ge Li. 2024. Generalization or memorization: Data contamination and trustworthy evaluation for large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 12039–12050.
- David Donoho and Jiashun Jin. 2004. Higher criticism for detecting sparse heterogeneous mixtures. *The Annals of Statistics*, 32(3):962–994.
- Michael Duan, Anshuman Suri, Niloofar Miresghallah, Sewon Min, Weijia Shi, Luke Zettlemoyer, Yulia Tsvetkov, Yejin Choi, David Evans, and Hannaneh Hajishirzi. 2024. Do membership inference attacks work on large language models? In *Proceedings of the Conference on Language Modeling (COLM)*.
- Yujuan Fu, Özlem Uzuner, Meliha Yetisgen, and Fei Xia. 2025. Does data contamination detection work (well) for LLMs? a survey and evaluation on detection assumptions. In *Findings of the Association for Computational Linguistics: NAACL 2025*.
- Shahriar Golchin and Mihai Surdeanu. 2025. Data contamination quiz: A tool to detect and estimate contamination in large language models. *Transactions of the Association for Computational Linguistics*, 13:809–830.
- Dirk Groeneveld, Iz Beltagy, Evan Walsh, Akshita Bhagia, Rodney Kinney, Oyvind Tafjord, Ananya Jha, Hamish Ivison, Ian Magnusson, Yizhong Wang, Shane Arora, David Atkinson, Russell Authur, Khyathi Chandu, Arman Cohan, Jennifer Dumas, Yanai Elazar, Yuling Gu, Jack Hessel, and 24 others. 2024. [OLMo: Accelerating the science of language models](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15789–15809, Bangkok, Thailand. Association for Computational Linguistics.
- Yuri I. Ingster. 1997. Some problems of hypothesis testing leading to infinitely divisible distributions. *Mathematical Methods of Statistics*, 6:47–69.
- Lucien Le Cam. 1986. *Asymptotic Methods in Statistical Decision Theory*. Springer.
- Inbal Magar and Roy Schwartz. 2022. Data contamination: From memorization to exploitation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 1578–1596.
- Pratyush Maini, Hengrui Jia, Nicolas Papernot, and Adam Dziedzic. 2024. LLM dataset inference: Did you train on my dataset? In *Advances in Neural Information Processing Systems*, volume 37.
- Justus Mattern, Fatemehsadat Miresghallah, Zhijing Jin, Bernhard Schölkopf, Mrinmaya Sachan, and Taylor Berg-Kirkpatrick. 2023. Membership inference attacks against language models via neighbourhood comparison. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 11330–11343.
- Andreas Maurer and Massimiliano Pontil. 2009. Empirical bernstein bounds and sample variance penalization. In *Proceedings of the 22nd Annual Conference on Learning Theory (COLT)*.
- Matthieu Meeus, Igor Shilov, Shubham Jain, Manuel Faysse, Marek Rei, and Yves-Alexandre de Montjoye. 2025. SoK: Membership inference attacks on LLMs are rushing nowhere (and how to fix it). In *IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*, pages 385–401.
- Yonatan Oren, Nicole Meister, Niladri S. Chatterji, Faisal Ladhak, and Tatsunori B. Hashimoto. 2024. Proving test set contamination in black-box language models. In *Proceedings of the International Conference on Learning Representations (ICLR)*.
- Haritz Puerto, Martin Gubri, Sangdoo Yun, and Seong Joon Oh. 2025. Scaling up membership inference: When and how attacks succeed on large language models. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 4165–4182.
- Harish Ramaswamy, Clayton Scott, and Ambuj Tewari. 2016. Mixture proportion estimation via kernel embeddings of distributions. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, pages 2052–2060.
- Walter J. Rogan and Beth Gladen. 1978. Estimating prevalence from the results of a screening test. *American Journal of Epidemiology*, 107(1):71–76.
- Vinay Samuel, Yue Zhou, and Henry Peng Zou. 2025. Towards data contamination detection for modern large language models: Limitations, inconsistencies, and oracle challenges. In *Proceedings of the 31st International Conference on Computational Linguistics (COLING)*.
- Clayton Scott. 2015. A rate of convergence for mixture proportion estimation, with application to learning from noisy labels. In *Proceedings of the 18th International Conference on Artificial Intelligence and Statistics (AISTATS)*, pages 838–846.
- Weijia Shi, Anirudh Ajith, Mengzhou Xia, Yangsibo Huang, Daogao Liu, Terra Blevins, Danqi Chen, and Luke Zettlemoyer. 2024. Detecting pretraining data from large language models. In *Proceedings of the International Conference on Learning Representations (ICLR)*.
- Aad W. van der Vaart. 1998. *Asymptotic Statistics*. Cambridge University Press.

Han Wang, Haoyu Li, Brian Ko, and Huan Zhang. 2026a. On the fragility of benchmark contamination detection in reasoning models. *arXiv preprint arXiv:2510.02386*.

Jeffrey G. Wang, Jason Wang, Marvin Li, and Seth Neel. 2026b. CheckMIABench: Firm foundations for membership inference attacks on language models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL, Short Papers)*, pages 364–370.

Roy Xie, Junlin Wang, Ruomin Huang, Minxing Zhang, Rong Ge, Jian Pei, Neil Zhenqiang Gong, and Bhuwan Dhingra. 2024. ReCaLL: Membership inference via relative conditional log-likelihoods. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 8671–8689.

Wojciech Zarzecki, Jan Dubiński, and Sebastian Cygert. 2026. The reliability gap in benchmark auditing: Distribution shift and scale as failure modes of contamination detection. In *Proceedings of the European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD)*.

Huixuan Zhang, Yun Lin, and Xiaojun Wan. 2024. PaCoST: Paired confidence significance testing for benchmark contamination detection in large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 1794–1809.

Jingyang Zhang, Jingwei Sun, Eric Yeats, Yang Ouyang, Martin Kuo, Jianyi Zhang, Hao Frank Yang, and Hai Li. 2025. Min-K%++: Improved baseline for detecting pre-training data from large language models. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Zheng Zhang, Qi Liu, Siyuan Liang, Ning Li, Zirui Hu, Weibo Gao, Rui Li, Zhenya Huang, Leszek Rutkowski, Baosheng Yu, and Dacheng Tao. 2026. Controllable contamination detection for reliable LLM evaluation with statistical guarantees. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 30122–30143.

A Proofs

A.1 Proof of the finite-sample power limit

Put $g = r - 1$. The one-observation likelihood ratio is $L_\alpha = 1 + \alpha g$. Because $\mathbb{E}_0 g = 0$ and $\mathbb{E}_0 g^2 = \rho^2$,

$$\mathbb{E}_0 L_\alpha^2 = 1 + \alpha^2 \rho^2.$$

The product likelihood ratio is $L_\alpha^{(m)} = \prod_{i=1}^m L_\alpha(Y_i)$. Independence therefore gives

$$\begin{aligned} 1 + \chi^2(Q_\alpha^{\otimes m} \| P_0^{\otimes m}) &= \mathbb{E}_0 \{L_\alpha^{(m)}\}^2 \\ &= \prod_{i=1}^m \mathbb{E}_0 L_\alpha(Y_i)^2 \\ &= (1 + \alpha^2 \rho^2)^m. \end{aligned} \quad (9)$$

For two simple hypotheses, every randomized test satisfies

$$a_m + b_m \geq 1 - \text{TV}(P_0^{\otimes m}, Q_\alpha^{\otimes m}).$$

Indeed,

$$\begin{aligned} a_m + b_m &= 1 - \{\mathbb{E}_{Q_\alpha^{\otimes m}} \phi_m - \mathbb{E}_{P_0^{\otimes m}} \phi_m\} \\ &\geq 1 - \text{TV}(P_0^{\otimes m}, Q_\alpha^{\otimes m}), \end{aligned}$$

because $0 \leq \phi_m \leq 1$. Cauchy–Schwarz applied to $\text{TV}(P, Q) = \frac{1}{2} \mathbb{E}_Q |dP/dQ - 1|$ gives

$$\text{TV}(P, Q) \leq \frac{1}{2} \sqrt{\chi^2(P \| Q)}.$$

Combining this inequality with eq. (9) and the trivial bound $\text{TV} \leq 1$ proves eq. (4).

Finally,

$$\begin{aligned} \pi_m(\alpha) - a_m &= \mathbb{E}_{Q_\alpha^{\otimes m}} \phi_m - \mathbb{E}_{P_0^{\otimes m}} \phi_m \\ &\leq \text{TV}(P_0^{\otimes m}, Q_\alpha^{\otimes m}). \end{aligned}$$

If $a_m \leq \tau$, substitution of the same total-variation bound gives eq. (5). $\square$

A.2 Statement and proof for a fixed contaminated subset

Lemma 6 (A fixed unknown contaminated subset). *Suppose exactly k of the m audited items are contaminated. For $S \subseteq [m]$, $|S| = k$, let $P_S = \bigotimes_{i \in S} P_1 \otimes \bigotimes_{i \notin S} P_0$, and let $\bar{b}_{m,k}$ be the type-II error averaged uniformly over such subsets. Then every test satisfies*

$$a_m + \bar{b}_{m,k} \geq 1 - \min\{1, C_{m,k}^{\text{fix}}\}, \quad (10)$$

$$\begin{aligned} C_{m,k}^{\text{fix}} &:= \frac{1}{2} \sqrt{\mathbb{E}(1 + \rho^2)^J - 1}, \\ \mathbb{E}(1 + \rho^2)^J &\leq \left(1 + \frac{k\rho^2}{m}\right)^k. \end{aligned} \quad (11)$$

where $J = |S \cap S'|$ for independent uniform k -subsets S, S' .

For achievability in a fixed regular channel, the local regime additionally requires $k/m \rightarrow 0$, $k\rho/\sqrt{m} \rightarrow h$, and $\text{Var}_1(r-1) < \infty$; there the symmetric oracle score $\sum_i (r(Y_i) - 1)$ attains the boundary without knowing the subset.

Let $\mathcal{S}_k = \{S \subseteq [m] : |S| = k\}$ and $\bar{P}_{m,k} = |\mathcal{S}_k|^{-1} \sum_{S \in \mathcal{S}_k} P_S$. Because type-II error is linear in the alternative,

$$\bar{b}_{m,k} = \mathbb{E}_{\bar{P}_{m,k}}(1 - \phi_m).$$

Under $P_0^{\otimes m}$, the likelihood ratio of P_S is $L_S = \prod_{i \in S} r(Y_i)$, so that of $\bar{P}_{m,k}$ is $\bar{L} = \mathbb{E}_S L_S$. Tonelli's theorem and independence give

$$\begin{aligned} 1 + \chi^2(\bar{P}_{m,k} \| P_0^{\otimes m}) &= \mathbb{E}_0 \bar{L}^2 \\ &= \mathbb{E}_{S,S'} \mathbb{E}_0 [L_S L_{S'}] \\ &= \mathbb{E}_{S,S'} (1 + \rho^2)^{|S \cap S'|}. \end{aligned}$$

Indeed, an index in $S \cap S'$ contributes $\mathbb{E}_0 r^2 = 1 + \rho^2$; every other selected index contributes $\mathbb{E}_0 r = 1$. Applying the testing-error/TV argument and the chi-square TV bound from section A.1 proves eq. (10).

It remains to prove eq. (11). Conditional on S , let $a_i = 1 + \rho^2$ for $i \in S$ and $a_i = 1$ otherwise. Uniformly sampling S' without replacement gives

$$\mathbb{E}[(1 + \rho^2)^J \mid S] = \binom{m}{k}^{-1} \sum_{\substack{T \subseteq [m] \\ |T|=k}} \prod_{i \in T} a_i.$$

Maclaurin's inequality bounds this normalized elementary symmetric mean by the k th power of the arithmetic mean:

$$\begin{aligned} \mathbb{E}[(1 + \rho^2)^J \mid S] &\leq \left(\frac{1}{m} \sum_{i=1}^m a_i \right)^k \\ &= \left(1 + \frac{k\rho^2}{m} \right)^k. \end{aligned}$$

The bound is independent of S , proving eq. (11). In particular, if $k^2 \rho^2 / m \rightarrow 0$, its right-hand side is at most $\exp(k^2 \rho^2 / m) \rightarrow 1$, so average testing power cannot exceed size asymptotically. Since $\sup_S b_m(S) \geq \bar{b}_{m,k}$, the same display is a minimax lower bound over fixed subsets.

For achievability in a fixed channel, let $Z_m = (\rho\sqrt{m})^{-1} \sum_i g(Y_i)$ and suppose $k/m \rightarrow 0$, $k\rho/\sqrt{m} \rightarrow h < \infty$, and $\text{Var}_1(g) < \infty$. Since

$$\mathbb{E}_1 g = \mathbb{E}_0 [r(r-1)] = \mathbb{E}_0 (r^2 - r) = \rho^2,$$

$\mathbb{E}_{P_S} Z_m = k\rho/\sqrt{m} \rightarrow h$. The centered sum over the $m - k$ clean indices, divided by $\rho\sqrt{m}$, converges to $\mathcal{N}(0, 1)$ by the ordinary CLT because $k/m \rightarrow 0$. The centered sum over the k member indices is $o_{P_S}(1)$ after the same normalization, since its variance is $k \text{Var}_1(g)/(\rho^2 m) \rightarrow 0$. Slutsky's theorem therefore gives $Z_m \rightsquigarrow \mathcal{N}(h, 1)$ under every P_S , while its null limit is $\mathcal{N}(0, 1)$. The one-sided score test consequently attains power $1 - \Phi(z_{1-\tau} - h)$ without knowing S . $\square$

A.3 Proof of the detection-boundary corollary

If $\alpha_m \rho_m \sqrt{m} \rightarrow 0$, then $m\alpha_m^2 \rho_m^2 \rightarrow 0$. Since $1 + x \leq e^x$ for $x \geq 0$,

$$C_m(\alpha_m, \rho_m) \leq \frac{1}{2} \sqrt{\exp(m\alpha_m^2 \rho_m^2) - 1} \rightarrow 0.$$

Apply eq. (5) and take the upper limit. $\square$

A.4 Local asymptotic optimality

Theorem 7 (Local asymptotic optimality; full statement). *Suppose $0 < \rho^2 < \infty$ and, for some $\epsilon > 0$, $\mathbb{E}_0 |g|^{2+\epsilon} < \infty$. Fix $h \geq 0$ and let $\alpha_m = h/(\rho\sqrt{m})$ for all sufficiently large m . Then*

$$\log \frac{dQ_{\alpha_m}^{\otimes m}}{dP_0^{\otimes m}} = hZ_m - \frac{h^2}{2} + o_{P_0}(1), \quad (12)$$

$$Z_m \rightsquigarrow \mathcal{N}(0, 1) \quad \text{under } H_0, \quad (13)$$

$$Z_m \rightsquigarrow \mathcal{N}(h, 1) \quad \text{under } H_{1,m}. \quad (14)$$

The score test $\mathbf{1}\{Z_m > z_{1-\tau}\}$ has asymptotic size τ and power $1 - \Phi(z_{1-\tau} - h)$. Every test sequence $\{\psi_m\}$ with $\limsup_m \mathbb{E}_0 \psi_m \leq \tau$ satisfies

$$\limsup_m \mathbb{E}_{\alpha_m} \psi_m \leq 1 - \Phi(z_{1-\tau} - h).$$

Proof. Let $g = r - 1$. Because $r \geq 0$, $g \geq -1$. Also,

$$\mathbb{E}_0 g = \int (p_1 - p_0) dy = 0, \quad \mathbb{E}_0 g^2 = \rho^2.$$

The central limit theorem gives

$$Z_m = \frac{1}{\rho\sqrt{m}} \sum_{i=1}^m g(Y_i) \rightsquigarrow \mathcal{N}(0, 1)$$

under $P_0^{\otimes m}$.

If $h = 0$, the likelihood ratio is identically one and all stated limits are immediate. Hence assume $h > 0$. For $\alpha_m = h/(\rho\sqrt{m})$, the log-likelihood ratio is

$$\Lambda_m = \sum_{i=1}^m \log(1 + \alpha_m g(Y_i)).$$

Let $\epsilon' = \min\{\epsilon, 1\}$. The moment condition and a union bound imply, for every fixed $c > 0$,

$$\begin{aligned} \mathbb{P}_0\left(\max_{i \leq m} |\alpha_m g(Y_i)| > c\right) \\ \leq m \mathbb{P}_0\left(|g| > \frac{c}{\alpha_m}\right) \\ \leq \frac{m \alpha_m^{2+\epsilon'}}{c^{2+\epsilon'}} \mathbb{E}_0 |g|^{2+\epsilon'} \rightarrow 0. \end{aligned}$$

On the event $\max_i |\alpha_m g_i| \leq 1/2$, Taylor's theorem gives a constant $C_{\epsilon'}$ such that

$$\left| \log(1+x) - x + \frac{x^2}{2} \right| \leq C_{\epsilon'} |x|^{2+\epsilon'}, \quad |x| \leq 1/2.$$

Therefore the summed remainder is bounded in probability by

$$\begin{aligned} C_{\epsilon'} \alpha_m^{2+\epsilon'} \sum_{i=1}^m |g(Y_i)|^{2+\epsilon'} \\ = O_{P_0}(m \alpha_m^{2+\epsilon'}) = o_{P_0}(1). \end{aligned}$$

The weak law of large numbers also gives $m^{-1} \sum_i g(Y_i)^2 \rightarrow \rho^2$ in P_0 probability. Combining these facts,

$$\begin{aligned} \Lambda_m &= \alpha_m \sum_{i=1}^m g(Y_i) \\ &\quad - \frac{\alpha_m^2}{2} \sum_{i=1}^m g(Y_i)^2 + o_{P_0}(1) \\ &= h Z_m - \frac{h^2}{2} + o_{P_0}(1), \end{aligned}$$

which is eq. (12).

Under $P_0^{\otimes m}$, (Z_m, Λ_m) converges jointly to $(Z, hZ - h^2/2)$ for $Z \sim \mathcal{N}(0, 1)$. Since $\mathbb{E}[\exp(hZ - h^2/2)] = 1$, Le Cam's third lemma applies and gives $Z_m \rightsquigarrow \mathcal{N}(h, 1)$ under $Q_{\alpha_m}^{\otimes m}$. Hence

$$\mathbb{P}_0(Z_m > z_{1-\tau}) \rightarrow \tau$$

and

$$\mathbb{P}_{\alpha_m}(Z_m > z_{1-\tau}) \rightarrow 1 - \Phi(z_{1-\tau} - h).$$

It remains to justify the power envelope. For each fixed m , the Neyman–Pearson lemma makes a possibly randomized threshold test of Λ_m most powerful for the simple alternative $Q_{\alpha_m}^{\otimes m}$. Let $\{\psi_m\}$ be any test sequence with $\limsup_m \mathbb{E}_0 \psi_m \leq \tau$. For every $\delta \in (0, 1 - \tau)$, it

eventually has size at most $s = \tau + \delta$, so Neyman–Pearson bounds its power by that of the likelihood-ratio test of size s . Under the null, eq. (12) implies $\Lambda_m \rightsquigarrow \mathcal{N}(-h^2/2, h^2)$, a continuous law. Hence the randomized NP threshold converges to $-h^2/2 + h z_{1-s}$. Contiguity and Le Cam's third lemma give $\Lambda_m \rightsquigarrow \mathcal{N}(h^2/2, h^2)$ under the alternative, so the NP power converges to $1 - \Phi(z_{1-s} - h)$. Taking the upper limit and then letting $\delta \downarrow 0$ gives the stated envelope. The score test attains it by the null and alternative limits above. $\square$

A.5 Proof of score efficacy

Let $\tilde{f} = f - \mathbb{E}_0 f$. Since $\mathbb{E}_0(r - 1) = 0$,

$$\begin{aligned} \Delta_f &= \int f(p_1 - p_0) dy \\ &= \mathbb{E}_0[(r - 1)\tilde{f}]. \end{aligned}$$

By Cauchy–Schwarz,

$$\Delta_f^2 \leq \mathbb{E}_0(r - 1)^2 \mathbb{E}_0 \tilde{f}^2 = \rho^2 \sigma_f^2.$$

Dividing by $\sigma_f^2 > 0$ proves $e_f \leq \rho$. Equality in Cauchy–Schwarz holds exactly when $\tilde{f} = c(r - 1)$ P_0 -almost surely for a nonzero constant c . Finally, linearity under the mixture gives

$$\begin{aligned} \mathbb{E}_{Q_\alpha} f - \mathbb{E}_0 f &= (1 - \alpha) \mathbb{E}_0 f + \alpha \mathbb{E}_1 f - \mathbb{E}_0 f \\ &= \alpha \Delta_f. \end{aligned}$$

The standard deviation of the sample mean under the null is $\sigma_f/\sqrt{m}$, so its standardized mean shift is $\alpha\sqrt{m} \Delta_f/\sigma_f$, with absolute value $\alpha\sqrt{m} e_f$. $\square$

A.6 Proof of the efficacy floor

For $z \in \{0, 1\}$, Hoeffding's inequality gives

$$\begin{aligned} \mathbb{P}\{|\bar{f}_z - \mu_z| > \varepsilon_{\text{cal}}(n_z)\} \\ \leq 2e^{-2n_z \varepsilon_{\text{cal}}(n_z)^2} = \frac{\delta_{\text{cal}}}{3}. \end{aligned}$$

For the clean sample, the pairwise sample variance of Maurer and Pontil (2009, Theorem 10) is exactly

$$\begin{aligned} V_{n_0} &= \frac{1}{n_0(n_0 - 1)} \sum_{i < j} \{f(Y_i^0) - f(Y_j^0)\}^2 \\ &= \hat{\sigma}_f^2, \end{aligned}$$

and its expectation is σ_f^2 . Their lower-tail standard-deviation inequality, with failure probability $\delta_{\text{cal}}/3$, yields

$$\begin{aligned} \mathbb{P}\{\sigma_f > \hat{\sigma}_f + u_{\text{cal}}(n_0)\} \\ \leq \frac{\delta_{\text{cal}}}{3}. \end{aligned}$$

A union bound therefore gives, with probability at least $1 - \delta_{\text{cal}}$, both mean bounds and $\sigma_f \leq \hat{\sigma}_f + u_{\text{cal}}(n_0)$. Popoviciu's deterministic inequality also gives $\sigma_f \leq 1/2$, hence $\sigma_f \leq \bar{\sigma}_f$. On the same event, the reverse triangle inequality gives $|\mu_1 - \mu_0| \geq \Delta_L$. Consequently,

$$e_f = \frac{|\mu_1 - \mu_0|}{\sigma_f} \geq \frac{\Delta_L}{\bar{\sigma}_f} = e_f,$$

and theorem 5 gives $\rho \geq e_f$. Replacing $\bar{\sigma}_f$ by the deterministic upper bound $1/2$ proves the variance-free alternative stated after the corollary. $\square$

A.7 Proof of the contamination certificate

Write $\mu_A = \mathbb{E}_{Q_\alpha} f$. Since $f \in [0, 1]$, two-sided Hoeffding bounds give, for any sample of size n and its corresponding mean μ ,

$$\mathbb{P}\{|\bar{f} - \mu| > \varepsilon_{\text{cert}}(n)\} \leq 2e^{-2n\varepsilon_{\text{cert}}(n)^2} = \frac{\delta_{\text{cert}}}{3}.$$

A union bound over the audit, clean-calibration, and seen-calibration samples shows that, with probability at least $1 - \delta_{\text{cert}}$, all three events

$$\begin{aligned} |\bar{f}_A - \mu_A| &\leq \varepsilon_{\text{cert}}(m), \\ |\bar{f}_0 - \mu_0| &\leq \varepsilon_{\text{cert}}(n_0), \\ |\bar{f}_1 - \mu_1| &\leq \varepsilon_{\text{cert}}(n_1) \end{aligned}$$

hold simultaneously. On this event,

$$N_L \leq \mu_A - \mu_0 = \alpha(\mu_1 - \mu_0)$$

by the mixture identity, while

$$D_U \geq \mu_1 - \mu_0.$$

Write $d = \mu_1 - \mu_0$. If $d \leq 0$, then $N_L \leq \alpha d \leq 0$, so $\underline{\alpha} = 0 \leq \alpha$ regardless of the sign of D_U . Now suppose $d > 0$. Then $D_U \geq d > 0$. If $N_L \leq 0$, again $\underline{\alpha} = 0 \leq \alpha$. If $N_L > 0$, then

$$\frac{N_L}{D_U} \leq \frac{\alpha d}{D_U} \leq \alpha.$$

The truncation to $[0, 1]$ preserves this inequality because $\alpha \in [0, 1]$. Thus $\underline{\alpha} \leq \alpha$ throughout the simultaneous event, which has probability at least $1 - \delta_{\text{cert}}$. $\square$

B The Contamination Certificate: Formal Statement

Theorem 8 (Finite-sample contamination certificate). *Fix any $f : \mathcal{Y} \rightarrow [0, 1]$ independently of all*

samples below. Let

$$\begin{aligned} \bar{f}_A &= \frac{1}{m} \sum_{i=1}^m f(Y_i^A), \\ \bar{f}_0 &= \frac{1}{n_0} \sum_{j=1}^{n_0} f(Y_j^0), \\ \bar{f}_1 &= \frac{1}{n_1} \sum_{k=1}^{n_1} f(Y_k^1), \end{aligned}$$

from mutually independent samples $Y_i^A \sim Q_\alpha$, $Y_j^0 \sim P_0$, and $Y_k^1 \sim P_1$. For confidence level $1 - \delta_{\text{cert}}$, define

$$\varepsilon_{\text{cert}}(n) = \sqrt{\frac{\log(6/\delta_{\text{cert}})}{2n}},$$

$$\begin{aligned} N_L &= (\bar{f}_A - \varepsilon_{\text{cert}}(m)) - (\bar{f}_0 + \varepsilon_{\text{cert}}(n_0)), \\ D_U &= (\bar{f}_1 + \varepsilon_{\text{cert}}(n_1)) - (\bar{f}_0 - \varepsilon_{\text{cert}}(n_0)), \end{aligned}$$

and

$$\underline{\alpha} = \begin{cases} \min\{1, \max\{0, N_L/D_U\}\}, & D_U > 0, \\ 0, & D_U \leq 0. \end{cases}$$

Then

$$\mathbb{P}_\alpha\{\underline{\alpha} \leq \alpha\} \geq 1 - \delta_{\text{cert}}.$$

C The Variance-Adaptive Efficacy Floor

Corollary 9 (A variance-adaptive finite-sample efficacy floor). *Fix a nonconstant $f : \mathcal{Y} \rightarrow [0, 1]$ independently of clean and seen calibration samples of sizes $n_0 \geq 2, n_1 \geq 1$. Let $\hat{\sigma}_f^2 = (n_0 - 1)^{-1} \sum_{j=1}^{n_0} (f(Y_j^0) - \bar{f}_0)^2$. For $\delta_{\text{cal}} \in (0, 1)$, define*

$$\begin{aligned} \varepsilon_{\text{cal}}(n) &:= \sqrt{\frac{\log(6/\delta_{\text{cal}})}{2n}}, \\ u_{\text{cal}}(n_0) &:= \sqrt{\frac{2 \log(3/\delta_{\text{cal}})}{n_0 - 1}}, \\ \bar{\sigma}_f &:= \min\{1/2, \hat{\sigma}_f + u_{\text{cal}}(n_0)\}, \\ \Delta_L &:= [|\bar{f}_1 - \bar{f}_0| - \varepsilon_{\text{cal}}(n_1) - \varepsilon_{\text{cal}}(n_0)]_+. \end{aligned}$$

Then, with probability at least $1 - \delta_{\text{cal}}$,

$$\begin{aligned} \rho &\geq e_f \geq \underline{e}_f, \\ \underline{e}_f &:= \frac{\Delta_L}{\bar{\sigma}_f}. \end{aligned} \tag{15}$$

The standard-deviation radius is the exact bounded-variable result of [Maurer and Pontil \(2009\)](#); it is useful when clean scores have low variance. Replacing $\bar{\sigma}_f$ by $1/2$ yields the fully variance-free but usually looser floor $2\Delta_L$. Either version is a score-efficacy lower bound and hence a certified floor on ρ ; neither pretends that a learned density ratio estimates ρ .

D The Matching Test in Detail

Write $g(y) = r(y) - 1$. The score at $\alpha = 0$ for the mixture family is g , and $\mathbb{E}_0 g = 0$, $\text{Var}_0(g) = \rho^2$. Define

$$Z_m = \frac{1}{\rho\sqrt{m}} \sum_{i=1}^m g(Y_i). \quad (16)$$

Local power envelope. Suppose $0 < \rho^2 < \infty$, $\mathbb{E}_0 |g|^{2+\epsilon} < \infty$ for some $\epsilon > 0$, and $\alpha_m = h/(\rho\sqrt{m})$. The full LAN statement in theorem 7 shows that Z_m converges to $\mathcal{N}(0, 1)$ under the null and $\mathcal{N}(h, 1)$ under the local alternative. The score test has asymptotic size τ and attains the power envelope

$$1 - \Phi(z_{1-\tau} - h), \quad (17)$$

which no asymptotic level- τ test exceeds. This is a local result, not a claim that one likelihood-ratio model is correct for every detector. It shows that the lower-bound scaling is attainable in each regular channel and gives the oracle asymptotic sample budget

$$m_{\text{oracle}} \approx \left(\frac{z_{1-\tau} + z_{1-\beta}}{\alpha\rho} \right)^2 \quad (18)$$

for size τ and power $1 - \beta$.

E Extensions

E.1 Adaptive transcripts

Proposition 10 (Adaptive information budget). *Let H_{t-1} be the transcript before round t , including all adaptively chosen prompts. Each round draws a fresh benchmark item; all turns involving one item are included in that round's single joint response, and no item is revisited later. Conditional on every history h , suppose the clean response law is $P_{0,t}(\cdot | h)$ and the contaminated law is*

$$Q_{\alpha,t}(\cdot | h) = (1 - \alpha)P_{0,t}(\cdot | h) + \alpha P_{1,t}(\cdot | h),$$

where

$$\chi^2(P_{1,t}(\cdot | h) \| P_{0,t}(\cdot | h)) \leq \rho_t^2$$

uniformly in h . Then every test of the final transcript satisfies

$$a_m + b_m \geq 1 - \min \left\{ 1, \left[\frac{1}{2} \sum_{t=1}^m \log(1 + \alpha^2 \rho_t^2) \right]^{1/2} \right\}.$$

Proof. Let $\mathbb{Q}_\alpha^{(m)}$ and $\mathbb{P}_0^{(m)}$ denote the two transcript laws, and define

$$\mathcal{K}_t(h) := \text{KL}(Q_{\alpha,t}(\cdot | h) \| P_{0,t}(\cdot | h)).$$

The chain rule for KL divergence gives

$$\text{KL}(\mathbb{Q}_\alpha^{(m)} \| \mathbb{P}_0^{(m)}) = \sum_{t=1}^m \mathbb{E}_{\mathbb{Q}_\alpha^{(m)}} [\mathcal{K}_t(H_{t-1})].$$

Conditionally on each history, direct expansion gives

$$\chi^2(Q_{\alpha,t}(\cdot | h) \| P_{0,t}(\cdot | h)) \leq \alpha^2 \rho_t^2.$$

For arbitrary $P \ll Q$, Jensen's inequality gives $\text{KL}(P \| Q) \leq \log\{1 + \chi^2(P \| Q)\}$. Thus the inner KL is at most $\log(1 + \alpha^2 \rho_t^2)$. Summing, applying Pinsker, and using the testing-error/TV inequality from section A.1 proves the claim. $\square$

E.2 Independent heterogeneous probes

If item i has clean and seen laws $(P_{0,i}, P_{1,i})$ with $\rho_i^2 = \chi^2(P_{1,i} \| P_{0,i})$, write $Q_{\alpha,i} = (1 - \alpha)P_{0,i} + \alpha P_{1,i}$. Independence gives the exact tensorization

$$\begin{aligned} R_m &:= 1 + \chi^2 \left(\bigotimes_{i=1}^m Q_{\alpha,i} \left\| \bigotimes_{i=1}^m P_{0,i} \right. \right) \\ &= \prod_{i=1}^m (1 + \alpha^2 \rho_i^2). \end{aligned}$$

Therefore the same chi-square/TV argument as theorem 3 gives

$$a_m + b_m \geq 1 - \min \left\{ 1, \frac{1}{2} \sqrt{R_m - 1} \right\}.$$

When $\max_i \alpha^2 \rho_i^2 = o(1)$, the cumulative local information is $\alpha^2 \sum_i \rho_i^2$. The corresponding heterogeneous score is $\sum_i (r_i(Y_i) - 1)$ normalized by $(\sum_i \rho_i^2)^{1/2}$. A Lindeberg condition yields the same Gaussian limit with $\rho^2 m$ replaced by $\sum_i \rho_i^2$.

E.3 Sampling a fixed finite benchmark

Suppose a benchmark has N distinct items and exactly k are contaminated. Evaluating all items is theorem 6 with $m = N$, so it does not require i.i.d. Bernoulli labels. Sampling $m < N$ distinct indices without replacement produces hypergeometric labels; the implementation conditions on the realized sample size and uses the exact randomization distribution (or a predeclared finite-population correction). Sampling indices with replacement recovers Bernoulli labels marginally but does not create new independent source items. Repeated decodes remain components of one joint item outcome.

F Validity Gates in Full

The four diagnostics summarized in §4.4:

1. **Blind separation.** A classifier that sees raw text but no target-model output must be at or near chance; otherwise content shift can masquerade as memorization (Das et al., 2025; Wang et al., 2026b). The operational rule, fixed in the run scripts before scoring: for same-corpus channels the gate passes if the AUC’s 95% interval contains 0.5 *or* the point AUC is at most the equivalence margin 0.55; for cross-dataset channels the content-shift limit is 0.75. One v3 channel (pythia-410m×wiki, AUC 0.544 [0.512, 0.579]) passes via the margin while its interval excludes exact chance—we state this rather than claiming compatibility with 0.5. For cross-dataset channels, where the pools differ by construction, the gate instead flags extreme content shift.
2. **Clean-channel transport.** Clean-control score distributions are compared across the clean and contaminated runs. Large shifts reject the no-spillover interpretation. Numerical rule as run in v5: two-sample KS statistic < 0.15 (observed range 0.046–0.103; per-channel values in table 6).
3. **Channel stability.** Conditional seen and clean score laws are compared across α using pre-specified two-sample distances. Scaling claims are limited to the range where these laws are stable. Numerical rule as run in v5: KS < 0.15 (observed range 0.034–0.094); the v5 size gate additionally requires the empirical-size Wilson lower bound ≤ 0.05 .
4. **Independence unit.** Near-duplicates, paraphrases, and repeated decodes are grouped by

source item. Effective m is the number of sampled item clusters.

G Probe Families, Baselines, and Statistical Protocol

G.1 Probe families and baselines

Gray-box (run in this paper). We evaluate mean loss and zlib-normalized loss (Carlini et al., 2021), Min-K% Prob (Shi et al., 2024), Min-K%++ (Zhang et al., 2025), and neighborhood loss (Matern et al., 2023). Each method produces one item-level score. Hyperparameters are fixed on the score-training split. The “best probe” reported anywhere in the paper is the probe with the highest calibration-split efficacy, selected before audit power is computed.

Gray-box (specified, not run). The protocol also admits ReCaLL (Xie et al., 2024) and a multifeature CAMIA-style score (Chang et al., 2025); neither is run in the reported experiments and no claim rests on them.

Black-box (specified, not run). The protocol specifies output correctness, sampling consistency, the Data Contamination Quiz (Golchin and Surdeanu, 2025), and canonical-order/exchangeability probes (Oren et al., 2024), with fixed model versions, prompts, decoding parameters, and retry policies, all generations for one item forming one joint probe. These are not run here; all reported results are gray-box.

Power-calibrated combination (specified, not run). The protocol’s intended primary score is a cross-fitted logistic density-ratio classifier over the baseline features available in the given access regime, with an identically tuned blind classifier on text-only features to expose confounding. It is not run in this paper; in every reported experiment the primary score is the calibration-selected best single probe defined above.

G.2 Validated-paraphrase pipeline

Candidate paraphrases of each injected question are generated by Qwen2.5-1.5B-Instruct (temperature 0.9, up to three attempts per item) and accepted only if (a) the answer string is preserved verbatim (deterministic check), (b) unigram Jaccard overlap with the original question is at most 0.6, and (c) two independently trained NLI judges, microsoft/deberta-large-mnli

and FacebookAI/roberta-large-mnli, both assign bidirectional entailment probability above 0.8; judge disagreements are rejected rather than adjudicated. Of 200 items, 138 obtained an accepted paraphrase (0.69). The rejection counts are per generation *attempt*, not per item: across 385 attempts (up to three per item), 175 failed both NLI judges, 70 split the judges, and 2 exceeded the overlap cap (247 rejected attempts + 138 accepted items = 385). Items without an accepted paraphrase are excluded from the paraphrase mechanism (never silently retained), and the paraphrase channel’s audits use only the 138 accepted items.

G.3 Provenance caveat in full

“Pile-trained” does not by itself certify that the base checkpoint never saw SQuAD, so we do *not* claim a clean-versus-contaminated contrast in the absolute sense. What the paired $\beta_{\text{train}=0}$ vs. $\beta_{\text{train}>0}$ continuations measure is the *incremental* exposure effect relative to a matched continuation with identical background, order, optimizer state, token budget, and initialization—a well-defined causal quantity even if the base model carries some prior exposure. Establishing absolute cleanliness would require post-cutoff or provably-absent benchmark material with exact provenance and near-duplicate removal, which we flag as the stronger design and leave to future work. Likewise, the surface-perturbation condition is an automatic lexical-overlap reducer, not a human-validated meaning-preserving paraphrase; we label it accordingly and do not claim semantic invariance.

G.4 Statistical protocol details

Probe counts are

$$m \in \{16, 32, 64, 128, 256, 512, 1024\},$$

truncated at the number of independent item clusters. The fresh clean-reference size n_0^{test} is 250 (half of the 500 held-out nonmembers), and each test reports m_{eff} . For each (α, m) , 500 Monte Carlo audit replicates are drawn only from the held-out audit pool. Rejection-rate intervals are Wilson intervals; model- and seed-level uncertainty is preserved by a hierarchical item/seed bootstrap. For every condition we report both the primary permutation-test rejection rate and $\mathbb{P}(\underline{\alpha} > 0)$, plus their difference; the latter is expected to be smaller because it uses the conservative certificate rule. In the full protocol the cross-fitted combination is the intended primary score in each access

regime; in the experiments reported here (which do not run the combination) the primary score is the calibration-selected best single probe, and the remaining probes are secondary and Holm-adjusted within channel.

The empirical onset $\hat{\alpha}_*(m)$ is the smallest isotonic-interpolated α whose lower power-interval endpoint exceeds 0.80 while the upper endpoint of the corresponding size interval is at most 0.07, a predeclared two-point tolerance above the nominal level. We fit

$$\log \hat{\alpha}_*(m) = a + b \log m$$

with a seed-clustered bootstrap interval for b . The theory predicts $b = -1/2$ only where $\hat{e}_f$ and the conditional channel remain stable. We compare predicted and observed $\log m_{\text{target}}$ using slope, intercept, R^2 , median multiplicative error, and calibration plots. The transport decision uses the predeclared criteria in section 4.4; a failed decision triggers the nontransport analysis rather than a post-hoc recalibration of the main claim.

H Secondary Scaling Diagnostic

After efficacy is frozen, rescaling by $\hat{e}_f$ and m_{eff} substantially collapses the held-out power curves onto the predicted information coordinate $\alpha \hat{e}_f \sqrt{m_{\text{eff}}}$: power correlates with this coordinate at $r = 0.68$ – 0.81 across probes (the “Power corr.” column of table 7). The detectability onset scales as $\alpha_*(m) \propto m^{-0.57}$, consistent with the theoretical $m^{-1/2}$ prediction of §3; mechanism-specific results are in table 12. We nonetheless report scaling as a diagnostic rather than the headline, because the collapse is approximate and the onset exponent, while close to $-1/2$, is estimated from a single model’s channel; the prospective power prediction is the primary, model-independent claim. Because the onset and $\hat{e}_f$ both arise from a standardized mean shift, agreement with $-1/2$ mainly checks whether the local approximation and channel-stability assumptions hold. It is not treated as independent evidence for the framework.

I Certificate Coverage at Audit Scale in Full

Two coverage numbers must not be conflated: the 99.0–99.5% and 0.938–0.953 figures in table 9 are endpoints of a *scaled-slack heuristic on unclipped scores*, not validation of the distribution-free certificate. The exact certificate of Theorem 8, on a

clipped $[0, 1]$ score frozen on a disjoint calibration half (measured margins $\hat{\Delta}_f = 0.11\text{--}0.26$), never fires at $m=512$ —zero false, zero nonzero certificates in all six channels; its 100% coverage is *vacuously satisfied*. Certifying $\alpha=0.1\text{--}0.2$ at measured margins needs balanced groups of $\sim 3,500\text{--}79,000$ items (§J; the v3 channels agree, table 3): a nonzero $\underline{\alpha}$ is trustworthy, a zero is uninformative, and at audit sizes the certificate is a null instrument, reported as a limitation.

The measured power onset and the analytic certificate resolution are distinct size-dependent thresholds (an earlier draft conflated them; corrected in table 8): the $n_0=250$ reference floors the resolution at ≈ 0.17 , so no audit-set size alone certifies $\alpha=0.1$, while the measured onset falls to ≈ 0.1 by $m=1,024$ in a pool-limited cell (full arithmetic in section J).

Scaling is a *diagnostic* only: power tracks $\alpha\hat{e}_f\sqrt{m_{\text{eff}}}$ ($r = 0.68\text{--}0.81$), onset slope -0.57 vs. the predicted $-1/2$; the collapse checks the local approximation, not the framework (section H, table 12).

J Power Onset vs. Certificate Resolution

An earlier draft conflated the measured power onset with the analytic certificate resolution $\alpha_{\star}^{\text{cert}}(m, n_0) = (\varepsilon_{\text{cert}}(m) + \varepsilon_{\text{cert}}(n_0))/\Delta_f$ and omitted the n_0 term; both are corrected in table 8. Even under the best-case bound $\Delta_f \leq 0.58$ (measured margins are $2\text{--}5\times$ smaller), the protocol’s $n_0=250$ reference floors the resolution at ≈ 0.17 : no audit-set size alone certifies $\alpha=0.1$; a balanced design needs $m=n_0 \approx 2,850$ best-case and $\approx 14,000\text{--}79,000$ at measured margins—the honest planning number. The measured onset falls to ≈ 0.1 by $m=1,024$, but that cell is bootstrap-resampled from ≤ 500 distinct held-out items with a plug-in clean reference, hence *pool-limited*—a secondary diagnostic, never an $n_0=250$ result (exact counts in the table 8 caption). The onset stays far below the certificate resolution at every size—the price of the orientation-free construction.

K Additional Result and Planning Tables

Tables 1–13 provide the supplementary per-seed injection results, validity-gate diagnostics, finite-sample planner evaluations, power-prediction results, certificate-resolution and coverage analyses, calibration-to-audit transport checks, sensitivity estimates, and empirical scaling diagnostics.

L Retrospective Pilot Details

The initial pilot used Pythia-1.4B with 400 labeled Pile members and 400 nominal nonmembers. It reported Cohen’s $d = 0.585$ and, after constructing post-hoc mixtures with $m \in \{4, 8, 16, 32, 64, 128, 256\}$, onsets

$$(.80, .60, .45, .30, .20, .15, .10),$$

whose log–log slope is -0.506 ($R^2 = 0.997$), reproduced from the committed pilot log (results_pythia/pythia_contamination_summary.json). All Monte Carlo mixtures reuse the same 800 controls, so their errors are correlated and the quoted R^2 is optimistic. Moreover, the mixture was formed after training, and the member/nonmember split was not protected against temporal, deduplication, and source confounding. The pilot is therefore an implementation check and planning illustration, not evidence for causal contamination or an independent validation of the scaling law.

M Reproducibility Protocol

M.1 Scope of this protocol

We do not release code or result artifacts with this submission. This appendix therefore specifies the experimental protocol at the level of detail required to re-implement it: model checkpoints and data sources, the split structure, the estimators and their tuning constants, the Monte-Carlo and bootstrap replicate counts, and the analysis decisions fixed before outcomes were computed. Every quantity reported in §6 is defined by the procedures below together with the constants tabulated in table 14 and the per-channel gate values in table 6.

M.2 Data and model specification

Reproduction requires: the base checkpoints (Pythia-70m/160m/410m/1.4b, GPT-Neo-125m/1.3B, GPT-2, OPT-125m, all from public releases), the corpora used to build channels (Wikipedia, CNN/DailyMail news, and SQuAD for the injection arm), and the Pile background corpus for continued pretraining. Channel construction is specified in §5: the v3 and v5 audits split one corpus into 700+700 member/nonmember halves and continue pretraining on the member half, so pools are text-exchangeable by construction and the split is reproducible from a seed alone.

Mechanism	seed 0	seed 1	seed 2	seed 3	seed 4
<i>Paired contrast D_s (common frozen probe: mean NLL), item-bootstrap 95% CI</i>					
Exact	0.90 [0.64,1.18]	1.05 [0.77,1.34]	1.03 [0.71,1.36]	0.99 [0.71,1.28]	1.07 [0.81,1.34]
Paraphrase	0.65 [0.39,0.92]	0.64 [0.37,0.92]	0.84 [0.51,1.18]	0.75 [0.50,1.01]	0.61 [0.37,0.88]
Surface	0.39 [0.20,0.58]	0.49 [0.27,0.71]	0.45 [0.24,0.66]	0.44 [0.23,0.65]	0.42 [0.24,0.61]
Answer-only	-0.02 [-0.17,0.14]	-0.08 [-0.26,0.11]	0.10 [-0.06,0.26]	-0.04 [-0.23,0.16]	-0.02 [-0.20,0.14]
<i>Paired contrast D_s (per-cell best probe), item-bootstrap 95% CI</i>					
Exact	0.90 [0.64,1.18]	1.22 [0.88,1.57]	1.15 [0.80,1.47]	0.99 [0.71,1.28]	1.07 [0.81,1.34]
Paraphrase	0.66 [0.35,0.99]	0.64 [0.37,0.92]	0.84 [0.51,1.18]	0.75 [0.50,1.01]	0.61 [0.37,0.88]
Surface	0.52 [0.28,0.77]	0.54 [0.29,0.77]	0.41 [0.21,0.60]	0.39 [0.21,0.57]	0.37 [0.22,0.54]
Answer-only	-0.02 [-0.17,0.14]	-0.08 [-0.26,0.11]	-0.83 [-3.96,0.95]	-0.04 [-0.23,0.16]	-0.20 [-0.89,0.29]
<i>Validity gate: empirical size at $\alpha=0$ (tolerance 0.07)</i>					
Exact	.021	.038	.001	.084	.043
Paraphrase	.106	.009	.033	.061	.080
Surface	.013	.015	.001	.004	.035
Answer-only	.306	.006	.000	.020	.103

Table 1: Per-seed five-seed paired injection results (Evaluation B). Paired contrast D_s = (exposed—held gap, contaminated) — (same gap, matched clean continuation) with item-bootstrap 95% intervals; the *common frozen probe* block (mean NLL, the modal calibration winner) is the basis of the ordering claim because the per-cell best probe (second block) differs across cells and mixes probes. On the common probe $D_s > 0$ with intervals excluding zero in all 15 exact/paraphrase/surface cells, and every answer-only interval covers zero. Adjacent seed-level paired differences on the common probe (mean $\pm$ sd over seeds; positive in 5/5 seeds each, sign-test $p = 2^{-5} = 0.031$, Holm-adjusted over the three comparisons $p_{\text{adj}} = 0.094$): exact—paraphrase 0.31 ± 0.12 , paraphrase—surface 0.26 ± 0.10 , surface—answer-only 0.45 ± 0.08 . (Best-probe adjacent differences, for reference: 0.36 ± 0.15 , 0.25 ± 0.14 , 0.68 ± 0.32 .) Bottom: empirical size of the exact permutation test at $\alpha=0$ per seed and mechanism (Wilson intervals in the released JSON); bold entries exceed the predeclared 0.07 tolerance and those channel—seed cells fail the size gate: exact/surface pass in 4/5 and 5/5 seeds, paraphrase and answer-only in 3/5—a further reason answer-only supports no detection claim.

M.3 Per-item record structure

Any re-implementation should retain, for every item, the fields that the analysis depends on:

- benchmark, split, and source-document cluster;
- membership provenance and training-stream position;
- deduplication and near-duplicate cluster identifiers;
- contamination mechanism, exposure count, and run/seed identifier;
- probe prompts, decoding settings, raw scores, and access regime;
- assignment to score-training, calibration, or audit split.

The cluster identifiers are load-bearing rather than bookkeeping: the effective m counts source-item clusters, so a re-implementation that treats near-duplicates or repeated decodes as independent probes will report inflated power at every budget.

M.4 Training controls

Table 14 summarizes the checkpoints, optimization settings, contamination construction, matched controls, stopping rule, and seeds used in the injection experiments.

M.5 Primary analysis decisions

Throughout the paper, “predeclared” means fixed in our run scripts and this protocol before the corresponding outcomes were computed; there is no public timestamped preregistration registry entry, so we do not use the term “pre-registered.”

1. The primary endpoint is benchmark-level power of the frozen one-sided permutation test at nominal size 0.05, not instance AUROC or $\underline{\alpha} > 0$.
2. The unit of independence is the source-item cluster.
3. Detector orientation and hyperparameters are selected without the audit split.
4. Settings failing the blind-separation or channel-stability gates are not pooled into the confirmatory scaling fit.

Channel	Blind AUC [CI]	Clean KS	Stab. KS	Clusters	Size [Wilson]	PASS	$m_{\text{pred}}/m_{\text{infl}}/m_{\text{obs}}$
pythia-410m×news	.953 [.941,.961]	.033	.056	1600	.017 [.011,.026]	F	meta
pythia-410m×wiki	.958 [.946,.969]	.034	.030	1600	.016 [.010,.025]	F	meta
pythia-1.4b×news	.953 [.940,.965]	.050	.041	1600	.040 [.030,.053]	F	meta
pythia-1.4b×wiki	.958 [.945,.968]	.057	.055	1600	.033 [.024,.044]	F	43/61/64
gpt-neo-1.3B×news	.953 [.939,.964]	.045	.054	1600	.009 [.005,.016]	F	72/101/256
gpt-neo-1.3B×wiki	.958 [.947,.966]	.064	.026	1600	.045 [.035,.058]	F	43/61/64

Table 2: Measured validity gates and locked transport validation for the six upgraded Evaluation A channels (best probe; exact permutation test). Every channel FAILS overall because the blind text-only AUC far exceeds the 0.75 content-shift limit—the pools come from different corpora—while the KS gates, cluster counts, and empirical size pass. The last column reports the frozen-inflation locked validation (§6.2): three channels are meta-calibration; on the three locked channels the inflated budget $m_{\text{infl}} = 1.40 \times m_{\text{pred}}$ is compared against the dyadic-grid onset m_{obs} , which only localizes the true onset to $(m_{\text{obs}}/2, m_{\text{obs}}]$. Reading with the censoring interval: the two 43/61/64 channels are *unresolved* ($61 \in (32, 64]$), the 72/101/256 channel is a *clear failure* ($101 \notin (128, 256]$)—so 1 failure, 2 censored, not 0/3 coverage.

Channel (v3)	Blind AUC [CI]	Size	Gates	$\hat{e}_f^{\text{cal}}/\hat{e}_f^{\text{audit}}$	$\hat{\Delta}_f$ (bal. m)	$m_{\text{pred}}/m_{\text{infl}}$	Power @ pred/infl
pythia-160m×wiki	.478 [.448,.512]	.003	PASS	2.41/1.79	.337 (8,411)	30/42	.67 [.60,.73] / .76 [.70,.81]
pythia-410m×wiki	.544 [.512,.579]	.000	PASS	4.62/4.31	.408 (5,753)	8/12	.56 [.49,.63] / .65 [.58,.71]
gpt-neo-125m×wiki	.493 [.457,.523]	.013	PASS	3.50/2.91	.415 (5,560)	14/20	.62 [.55,.68] / .725 [.66,.78]
gpt-neo-125m×squad	.510 [.479,.545]	.033	PASS	0.59/0.57	.153 (40,704)	—*	—
pythia-160m×squad	.544 [.513,.579]	.285	F (size)	0.46/0.58	.081 (145,362)	—	—
pythia-410m×squad	.489 [.461,.516]	.572	F (size)	0.52/0.49	.133 (54,214)	—	—

Table 3: v3 same-corpus continued-pretraining audit: random member/nonmember split of one corpus (700+700), members trained into the model ($k=4, 1,200$ steps), so pools are text-exchangeable by construction. Blind AUC sits at or near chance under the pre-specified rule (interval contains 0.5 or point ≤ 0.55 ; section F); 4/6 channels pass all gates and the two Pythia×SQuAD channels fail the *size* gate (bold) and are excluded from confirmatory analysis. $\hat{\Delta}_f$ is the measured clipped-score margin with the implied balanced $m=n_0$ for certifying $\alpha=0.1$ in parentheses; the exact certificate never fires at $m=100$ (coverage vacuously 1.0, zero nonzero certificates). The last column gives *direct, censoring-free* power at the frozen budgets m_{pred} and $m_{\text{infl}}=1.40 \times m_{\text{pred}}$ (200 trials, Wilson 95%): at m_{pred} all three intervals exclude the 0.8 target (three clear failures); at m_{infl} two exclude it and one is unresolved (0.76 [0.70, 0.81]). *Unattainable: $M_{\text{eff}} \geq n_0=250$.

5. The transport claim is withdrawn if either pre-declared prediction criterion in section 4.4 fails; no audit-split recalibration repairs that decision.
6. Every zero-contamination run is included in size and false-certificate estimates.
7. Missing or failed runs and all exclusions are reported by seed and condition.

N Related Work in Full

This section expands §7. We organize by what each line of work *delivers* rather than by method family, since the paper’s contribution is a deliverable (prospective power, and a lower bound on α) rather than a new detector or a new mixture bound.

N.1 Behavioral detectors

The dominant evidence channel for contamination is behavioral: a statistic of the model’s response to a benchmark item, compared against some reference. Loss and its compression-normalized vari-

ants (Carlini et al., 2021) are the baseline; Min-K% Prob (Shi et al., 2024) and Min-K%++ (Zhang et al., 2025) threshold low-probability token subsets; neighborhood comparison (Mattern et al., 2023) replaces an external reference with perturbed counterfactuals; ReCaLL (Xie et al., 2024) contrasts conditional log-likelihoods under prefixes; CAMIA-style features (Chang et al., 2025) exploit context-dependence of memorization. Black-box variants elicit exposure through canonical-order exchangeability (Oren et al., 2024) or direct recall quizzing (Golchin and Surdeanu, 2025).

In our framework each of these is a scalar score f , and Proposition 5 reduces it to a single number, the efficacy $e_f = |\mathbb{E}_1 f - \mathbb{E}_0 f|/\sigma_{f,0} \leq \rho$, which upper-bounds the local power of the oriented mean-score audit built from f . This is deliberately lossy: e_f does not summarize every test computable from f , and two detectors with equal AUROC can have different low- α efficacy. What the reduction buys is that detector comparison, channel comparison,

Channel (v4 planner)	Blind AUC [CI]	Size	Gates	Probe	$\hat{e}_f^{\text{cal}}$	m_{pred}	m_{plan}	Power @ pred	Power @ plan
pythia-160m×squad	.544 [.513,.579]	.302	F (size)	mink	0.46	—	—	—	—
pythia-160m×wiki	.547 [.517,.579] [†]	.000	PASS	zlib	1.20	190	abstain	.99 [.96,1.00]	—
pythia-410m×squad	.506 [.474,.535]	.008	PASS	neigh.	0.50	—*	abstain	—	—
pythia-410m×wiki	.538 [.508,.573] [†]	.000	PASS	mink	3.98	11	32	.625 [.56,.69]	.87 [.82,.91]
gpt-neo-125m×squad	.478 [.446,.513]	.023	PASS	mink	0.56	—*	abstain	—	—
gpt-neo-125m×wiki	.496 [.461,.529]	.025	PASS	mink	2.80	22	60	.79 [.73,.84]	.97 [.94,.99]

Table 4: Complete v4 finite-sample planner record (fresh same-corpus continued-pretraining channels; 700+700 members/nonmembers, calibration/audit halves of 350 clusters each, $\alpha_0=0.2$, $n_0=250$, 199-permutation add-one test, level 0.05; planner: $B=30$ outer item bootstraps, 60 inner Monte-Carlo trials per replicate, lower bound = 0.10 quantile, m -grid up to 192; evaluation: 200 fresh trials with Wilson 95% intervals). Best probe frozen by the calibration rule. m_{pred} is the Gaussian plug-in budget; m_{plan} the planner budget; “abstain” means the bootstrap lower bound never reached 0.80 on the grid—on pythia-160m×wiki the planner abstains while the large Gaussian budget succeeds ($m=190$: 0.99), a conservative failure reported straight. [†]Passes the pre-specified blind rule via the point margin (≤ 0.55) although the upper CI exceeds 0.55; under a proper equivalence gate (upper CI ≤ 0.55) these channels are excluded and the head-to-head rests on gpt-neo-125m×wiki alone. *Unattainable: $M_{\text{eff}} \geq n_0$. m_{plan} selects the smallest m from pointwise bootstrap bounds and is not selection-valid simultaneous inference (§6.2).

and budget planning become the same measurement. We run five gray-box families (§G); ReCaLL, CAMIA-style, and black-box probes are specified in the protocol but not run, and no reported result rests on them.

N.2 The reliability gap

A second literature evaluates detectors rather than proposing them, and reports a consistent picture of fragility. Duan et al. (2024) find membership inference near chance on LLMs at realistic scales; Fu et al. (2025) audit the assumptions detection methods rely on and find them frequently violated; Meeus et al. (2025) systematize the field’s evaluation failures; Samuel et al. (2025) document inconsistency across modern models and the absence of a usable oracle; Dekoninck et al. (2024a) show that contamination detection is cheap to evade; Wang et al. (2026a) extend this to reasoning models. Most directly, Zarzecki et al. (2026) run 25 models and record only 201 of 335 audit outcomes correct, with distribution shift producing false positives and post-hoc inference run at sample sizes too small to support its conclusions.

Our reading of this literature is that it diagnoses two distinct failures that are usually reported together: *invalidity* (the clean reference does not match the suspect channel, so separation reflects content rather than exposure) and *underpower* (the channel is informative but the audit was too small to resolve the exposed fraction). The first is what our gates (§4.4, Appendix F) test; the second is what efficacy and the planner (§6.2) quantify. A non-rejection that distinguishes these two states is

interpretable; a non-rejection that does not is the object the reliability-gap literature keeps finding. The locked external validation on the open 25-model release of Zarzecki et al. (2026)—efficacy calibrated without their audit outcomes, predicting their documented failures—is specified but not yet run, and we do not claim it.

N.3 Aggregation methods and their estimands

A third line lifts item-level scores to benchmark- or dataset-level decisions. LLM Dataset Inference (Maini et al., 2024) combines many membership features into a hypothesis test that a dataset was trained on. PaCoST (Zhang et al., 2024) constructs paired counterparts and tests for confidence asymmetry. Puerto et al. (2025) study when and how membership attacks succeed as a function of scale and data quantity. ConStat (Dekoninck et al., 2024b) estimates *performance inflation* by comparing a model’s benchmark score against a reference set of models on a matched benchmark. FTD (Zhang et al., 2026) supplies false-discovery-rate control over a set of contamination decisions.

These are complementary, and Table 15 makes the distinction explicit. A valid p -value certifies a rejection but is silent about which α the audit had power to detect—the quantity an auditor needs to interpret a non-rejection. Performance inflation measures how much a score is distorted, which depends on exposure fraction, mechanism, and the benchmark’s headroom jointly, and does not identify the fraction exposed. FDR control governs the error rate of a discovery set across many benchmarks and says nothing about the resolution of a

Channel (v5)	Blind AUC CI	Size	Gate	Probe	$\hat{e}_f^A$	m_{pred}	m_A	m_{plan}	$\hat{m}_*$	90% int.	$\frac{m_{\text{plan}}}{\hat{m}_*}$	$\frac{m_{\text{pred}}}{\hat{m}_*}$	Pow@pred	Pow@plan
pythia-70m×wiki	[.447,.521]	.040	PASS	neigh.	2.77	22	—	—	—	—	—	—	.01	—
pythia-70m×cnn	[.458,.513]	.047	PASS	zlib	2.24	36	56	56	56	[30,56]	1.00	0.64	.72	.79
pythia-160m×wiki	[.454,.514]	.052	PASS	zlib	1.67	72	120	120	84	[45,124]	1.43	0.86	.74	.89
pythia-160m×cnn	[.498,.557]	.048	F (equiv.)	mink	2.00	46	68	102	44	[34,72]	2.32	1.05	.78	.91
pythia-410m×wiki	[.495,.563]	.050	F (equiv.)	mink	3.59	13	28	42	16	[16,29]	2.63	0.81	.69	.94
pythia-410m×cnn	[.463,.525]	.057	PASS	mink	2.60	26	52	52	44	[24,52]	1.18	0.59	.62	.79
gpt-neo-125m×wiki	[.482,.540]	.032	PASS	mink	3.15	17	32	48	30	[22,33]	1.60	0.57	.61	.90
gpt-neo-125m×cnn	[.476,.543]	.043	PASS	zlib	2.85	21	36	54	38	[22,40]	1.42	0.55	.65	.86
gpt2×wiki	[.488,.549]	.038	PASS	mink	2.39	31	52	104	52	[30,52]	2.00	0.60	.69	.92
gpt2×cnn	[.467,.529]	.055	PASS	mink	1.88	54	88	88	68	[44,92]	1.29	0.79	.71	.83
opt-125m×wiki	[.480,.547]	.060	PASS	mink	3.47	14	30	30	26	[18,31]	1.15	0.54	.67	.84
opt-125m×cnn	[.499,.561]	.053	F (equiv.)	mink	3.07	18	38	57	36	[22,38]	1.58	0.50	.64	.92

Table 5: Complete v5 planner record: 12 same-corpus continued-pretraining channels (700+700 members/nonmembers, $k=4$, 1,200 steps; item splits cal-A/cal-B/audit = 175/175/350 clusters; $\alpha_0=0.2$, $n_0=250$, 199-permutation add-one test at level .05). Gate: predeclared one-sided equivalence rule (blind AUC upper 95% CI ≤ 0.55) plus KS and size gates, 9/12 pass; the stricter two-sided sensitivity gate (entire CI in $[0.45, 0.55]$) passes 8/12, additionally excluding pythia-70m×wiki; failures marked, excluded from endpoint aggregates; per-channel KS statistics and size Wilson intervals in table 6. Probe and standardization frozen on cal half A ($\hat{e}_f^A$ = cal-A efficacy). Stage A: $B=30$ item bootstraps $\times$ 60 Monte-Carlo trials per replicate at each grid m (grid to 192); $m_A = \min\{m : 0.10\text{-quantile of replicate power} \geq 0.80\}$; the 90% $\hat{m}_*$ interval is the $[q_{.05}, q_{.95}]$ of per-replicate minimal budgets and is *descriptive* ($B=30$; probe selection not re-run). Stage B (rung-selection-valid conditional on the empirical cal-B distribution; cal-B’s 175 clean clusters are resampled with replacement to reach $n_0=250$, so score-law uncertainty, cluster reuse, and cal-to-audit transport are outside the guarantee): pre-specified ladder $\{m_A, [1.5m_A], 2m_A\}$ on untouched cal half B, 300 trials each, one-sided Bonferroni-corrected ($\alpha=0.10/3$) lower bounds; m_{plan} = smallest rung clearing 0.80; escalation occurred in 3/8 channels. Audit half: a dense power curve gives the *estimated* onset $\hat{m}_*$ (200 trials per rung; a noisy threshold, not ground truth—e.g. pythia-70m×cnn has $m_{\text{plan}} = \hat{m}_* = 56$ with $\hat{\pi}(56) = 0.785$); Pow@pred/Pow@plan are fresh 200-trial powers at m_{pred} (Gaussian plug-in from cal A) and m_{plan} . On pythia-70m×wiki stage A never certified 0.80 (abstention; 90% interval censored at the grid edge) while the Gaussian budget delivers 0.01 power—the cal-selected *neighborhood* probe does not transport, and the planner’s abstention, not the plug-in’s prescription, is the correct behavior. Aggregates over gate-passing channels: Gaussian clear failures 9/9 ($m_{\text{pred}}/\hat{m}_*$ median 0.59 over the 8 channels with defined $\hat{m}_*$); planner covered 6/8, two unresolved, zero clear failures, conservatism median 1.36, interval coverage 8/8 (descriptive, subject to the $\hat{m}_*$ estimation noise above).

single audit. Our two deliverables are the prospective statement (what budget is required, and what a given budget can detect) and the retrospective one (a distribution-free lower confidence bound on α itself). The paper’s main negative result is that the first is harder to deliver than the local theory suggests: efficacy transports, but its Gaussian inversion into a budget does not (§6.2).

N.4 Validity: blind baselines and matched construction

Das et al. (2025) show that classifiers seeing only the raw text, with no access to the target model, match or exceed membership inference attacks on foundation models—the separation being audited is often a property of the pools, not of the model. Wang et al. (2026b) formalize the corresponding construction requirement, drawing members and nonmembers from windows matched on source, length, packing, deduplication, and stream position around a training checkpoint. Fu et al. (2025) reach the same conclusion from the assumptions side.

We adopt this as a precondition with numerical thresholds fixed in the run scripts before scoring, rather than as a discussion-section caveat, and we report gate values for every channel including the failures. All six upgraded Evaluation A channels fail blind separation at text-only AUC ≈ 0.95 , because their member and nonmember pools come from different corpora; we report them as a demonstration that the gate works, and rest the confirmatory reading on the same-corpus continued-pretraining channels, where blind AUC sits at 0.48–0.54. The stricter two-sided equivalence form of the gate (entire AUC interval within $[0.45, 0.55]$) is reported as a sensitivity analysis in Table 6 and excludes exactly the channel where the planner abstains.

N.5 Sparse-mixture detection

The statistical content of the $\alpha\rho\sqrt{m}$ boundary is classical. Ingster (1997) and Donoho and Jin (2004) establish detection boundaries for sparse heterogeneous mixtures and the higher criticism statistic;

Channel	Blind AUC 95% CI	1-sided gate	2-sided gate	Clean-shift KS	Stability KS	Empirical size [Wilson 95%]
pythia-70m×wiki	[0.447, 0.521]	PASS	fail	0.069	0.071	0.040 [0.027, 0.059]
pythia-70m×cnn	[0.458, 0.513]	PASS	PASS	0.066	0.066	0.047 [0.032, 0.067]
pythia-160m×wiki	[0.454, 0.514]	PASS	PASS	0.060	0.060	0.052 [0.037, 0.072]
pythia-160m×cnn	[0.498, 0.557]	fail	fail	0.046	0.051	0.048 [0.034, 0.069]
pythia-410m×wiki	[0.495, 0.563]	fail	fail	0.086	0.046	0.050 [0.035, 0.070]
pythia-410m×cnn	[0.463, 0.525]	PASS	PASS	0.063	0.051	0.057 [0.041, 0.078]
gpt-neo-125m×wiki	[0.482, 0.540]	PASS	PASS	0.077	0.074	0.032 [0.020, 0.049]
gpt-neo-125m×cnn	[0.476, 0.543]	PASS	PASS	0.066	0.051	0.043 [0.030, 0.063]
gpt2×wiki	[0.488, 0.549]	PASS	PASS	0.083	0.089	0.038 [0.026, 0.057]
gpt2×cnn	[0.467, 0.529]	PASS	PASS	0.103	0.034	0.055 [0.039, 0.076]
opt-125m×wiki	[0.480, 0.547]	PASS	PASS	0.060	0.094	0.060 [0.044, 0.082]
opt-125m×cnn	[0.499, 0.561]	fail	fail	0.089	0.094	0.053 [0.038, 0.074]

Table 6: Per-channel v5 gate values (as-run thresholds, all auditable from the released JSON): blind AUC upper 95% CI ≤ 0.55 (predeclared one-sided equivalence form), clean-transport KS < 0.15 , channel-stability KS < 0.15 , empirical-size Wilson lower bound ≤ 0.05 (n_{size} trials at $\alpha=0$). The one-sided gate passes 9/12; the stricter two-sided sensitivity gate (entire AUC CI within $[0.45, 0.55]$) passes 8/12, additionally excluding pythia-70m×wiki (lower CI $0.447 < 0.45$: possible reverse-direction separation)—the same channel where the planner abstains. The as-run size rule has the *detection* direction (a channel fails only when inflation is conclusively detected); a *size-equivalence* rule demanding the Wilson upper bound ≤ 0.07 (the paper’s declared size tolerance) is stricter, and combining it with the two-sided AUC gate leaves 4/12 channels clearly passing all gates (pythia-70m×cnn, gpt-neo-125m×wiki/cnn, gpt2×wiki); on those four the planner endpoint reads 3/4 at target, one unresolved, zero clear failures. No KS value approaches the 0.15 threshold (max 0.103); every size interval covers the nominal 0.05.

Probe	Power R^2	Power MAE	Power corr.	Slack cov.
Mean NLL	0.813	0.101	0.726	0.995
zlib-normalized NLL	0.931	0.066	0.675	0.990
Min-K%	0.986	0.029	0.706	0.990
Min-K%++	0.833	0.089	0.788	0.995
Neighborhood	0.992	0.023	0.811	0.995

Table 7: Prospective prediction (primary endpoint): audit power predicted from the *frozen calibration efficacy* matches held-out power across the tested (α, m) grid with $R^2 = 0.81$ –0.99 and mean absolute power error 0.02–0.10; the correlation of held-out power with the theoretical coordinate $\alpha \hat{e}_f \sqrt{m}$ is 0.68–0.81. The last column is the *heuristic scaled-slack* interval’s empirical coverage on unclipped scores (§6.3), a measured diagnostic of that heuristic—not coverage of the distribution-free prevalence certificate, which never fires at these sample sizes. R^2 alone is not claimed to establish calibration; the MAE column bounds systematic prediction bias directly.

Cai et al. (2011) extend to heteroscedastic mixtures; Cai and Wu (2014) give optimal detection against a given null distribution. The local asymptotic machinery we use for the matching test—LAN expansions, Le Cam’s third lemma, and the power envelope—is standard (Le Cam, 1986; van der Vaart, 1998).

We claim no new mixture boundary. Theorem 3 is an exact second-moment calculation for the specific channel $Q_\alpha = (1 - \alpha)P_0 + \alpha P_1$, sharpened only in that it passes through χ^2 and total variation rather than KL and Pinsker, and Lemma 6 extends it to the fixed-unknown-subset design an auditor actually faces. What is new here is the identifica-

tion: $\rho^2 = \chi^2(P_1 \| P_0)$ is not a nuisance parameter but the audit-observable channel quantity, estimable from matched controls and lower-bounded with finite-sample certainty (Corollary 9), and it replaces the informal capacity-based arguments that appear in prior contamination discussions—capacity alone does not upper-bound ρ without a training-stability theorem.

N.6 Prevalence estimation

Our certificate is a distribution-free relative of classical prevalence estimation. Rogan and Gladen (1978) correct an observed screening-test positive rate by known sensitivity and specificity, which is the population version of the ratio in Theorem 8. The modern learning-theoretic treatment is mixture proportion estimation from positive and unlabeled data (Blanchard et al., 2010; Scott, 2015; Ramaswamy et al., 2016), which achieves better rates under distributional assumptions such as irreducibility or separability.

We deliberately give those rates up. Our bound holds for any bounded score learned on independent data, with no orientation assumption—a reversed score returns $\underline{\alpha} = 0$ rather than a spurious positive—and no assumption on the score’s law beyond boundedness. The price is quantified in Appendix J and is severe: with an $n_0=250$ reference the resolution floors at ≈ 0.17 , and certifying $\alpha = 0.1$ at measured margins requires balanced groups of 14,000–79,000 items. The cer-

Table 8: Measured power onset vs. analytic certificate resolution on pythia-1.4b ($\hat{e}_f = 1.16$, best-case $\Delta_f \leq 0.58$, $\delta_{\text{cert}} = 0.05$, $\varepsilon_{\text{cert}}(n) = \sqrt{\log(6/\delta_{\text{cert}})/(2n)}$ from Theorem 8). Certificate rows are population-margin *planning thresholds under the margin upper bound*, not measured onsets: the measured $\hat{\Delta}_f$ of the clipped audit score (0.11–0.26 across the up-graded channels) can only be smaller, which makes every threshold larger: under the measured margins the balanced $\alpha=0.1$ crossing moves from $\approx 2,850$ to $\approx 14,000$ –79,000 per group. The protocol row fixes $n_0=250$, giving 0.23 at $m=2,000$, 0.21 at $m=4,000$, and floor $\varepsilon_{\text{cert}}(250)/\Delta_f \approx 0.17$ as $m \rightarrow \infty$; the balanced row is a *hypothetical* design with $m=n_0$, reaching 0.1 near $m=n_0 \approx 2,850$. The power onset is the measured 0.8-power fraction with fitted slope $m^{-0.57}$ over $m \leq 1,024$, matching the $m^{-1/2}$ law. Exact design counts for this initial-audit onset, disclosed in full: the channel is 500+500 items, split 250/250 per class into calibration and held-out halves; audit sets are bootstrap-resampled *with replacement* from the ≤ 500 distinct held-out items, and the test is a standardized-mean threshold $1.645/\sqrt{m}$ with the held-out clean half’s (μ_0, σ_0) plugged in as known (no finite-reference penalty). At $m=1,024$ the audit set therefore contains at most 500 distinct items: the $\alpha=0.1$ onset is *pool-limited and secondary*, and must not be read as 1,024 independent items under any convention. The confirmatory measurements are the exact-permutation audits (tables 2 and 3), which draw without replacement and cap m at the independent-cluster count.

m	256	512	1024	2000	4000
$\alpha_{\star}^{\text{cert}}(n_0=250, \text{protocol})$	0.34	0.29	0.25	0.23	0.21
$\alpha_{\star}^{\text{cert}}(n_0=m, \text{hypothetical})$	0.33	0.24	0.17	0.12	0.08
$\alpha_{\star}^{\text{pow}}(\text{test power } 0.8, \text{measured})$	0.2	0.2	0.1	—	—

tificate never fires at our audit sizes. Whether an assumption-bearing MPE estimator can supply a usable interval at realistic benchmark sizes, with coverage that survives the gates, is the natural next question and we do not answer it.

N.7 Power analysis in evaluation design

Sample-size planning is routine in the empirical sciences and rare in NLP evaluation, where benchmark size is typically inherited from the dataset rather than chosen to achieve a target power against a stated alternative. The contamination setting makes the omission costly, because the alternative of interest is small (α of a few percent) and the per-item signal is weak, so the regime is exactly the one where an unplanned audit returns an uninterpretable non-rejection. Our finding that the textbook Gaussian sample-size formula is miscalibrated at the budgets it prescribes (§6.2) is a cau-

Setting	Certificate coverage	Med. gap	Best-probe e_f
Checkpoint mixture (Eval. A)	0.990–0.995	0.13–0.16	1.16
Exact injection (Eval. B)	0.951	0.115	0.741
Surface perturbation (Eval. B)	0.938	0.191	0.343
Answer-only (Eval. B)	0.953	0.200	0.150

Table 9: *Heuristic* scaled-slack lower-bound coverage at true $\alpha = 0.2$ —not the exact certificate of Theorem 8, whose validation this table does not provide (see §6)—across the Evaluation A post-hoc checkpoint mixture (pythia-1.4b) and the Evaluation B causal injections (continued-pretraining on pythia-160m under three mechanisms; best-probe coverage pooled over three training seeds, 150 Monte-Carlo trials per seed). Pooled counts and Wilson 95% intervals: exact 428/450 = 0.951 [0.927, 0.967] (per-seed 0.980, 0.947, 0.927); surface 422/450 = 0.938 [0.912, 0.957] (0.933, 0.933, 0.947); answer-only 429/450 = 0.953 [0.930, 0.969] (0.960, 0.947, 0.953). Every interval contains the nominal 0.95, so the sub-nominal point values are statistically compatible with Monte-Carlo variation; we do not attribute them to any channel property. The completed runs implement the certificate with a scaled-slack approximation on unclipped scores rather than the exact clipped-[0, 1] construction of Theorem 8, so empirical coverage is a measured endpoint here, not a by-construction guarantee. Detectability falls sharply and stably with the injection mechanism—exact ($\hat{e}_f = 0.741$; per-seed 0.645, 0.862, 0.716) > surface perturbation (0.343; 0.360, 0.355, 0.313) > answer-only (0.150; 0.119, 0.152, 0.178), an ordering that holds in all three seeds—exactly the surface-form dependence the channel predicts.

tion that generalizes past contamination: when a planning equation prescribes small m , the approximation licensing it is at its worst, and the budget should be obtained by simulating the deployed test rather than by inverting a closed form.

Score	$\hat{e}_f^{\text{cal}}$	$M_{\text{eff}}^{\text{cal}}$	$m_{\text{pred}}^{\text{cal}}$	$\hat{e}_f^{\text{test}}$	$M_{\text{eff}}^{\text{test},\dagger}$	eff.-transp.
Loss (mean NLL)	1.496	69.0	96	0.859	209.2	$3.03\times$
zlib-norm. loss	0.774	258.1	—	1.154	116.0	$2.23\times$
Min-K%	0.896	192.5	837	0.982	160.3	$1.20\times$
Min-K%++	0.976	162.1	461	0.598	432.3	$2.67\times$
Neighborhood	0.519	572.5	—	0.512	590.5	$1.03\times$

Table 10: Evaluation A calibration-to-audit transport on EleutherAI/pythia-1.4b (Pile members vs. held-out nonmembers, 500+500 items). $\hat{e}_f^{\text{cal}}$ is the *frozen calibration* efficacy; $M_{\text{eff}}^{\text{cal}} = (2.486/(\alpha_0 \hat{e}_f^{\text{cal}}))^2$ is the prospective effective sample size at $\alpha_0 = 0.2$, size .05, power .8; $m_{\text{pred}}^{\text{cal}} = \lceil M_{\text{eff}}^{\text{cal}} n_0 / (n_0 - M_{\text{eff}}^{\text{cal}}) \rceil$ is the required suspect-set size under a *finite* clean reference of $n_0=250$ items (“—” when $M_{\text{eff}}^{\text{cal}} \geq n_0$, i.e. *unattainable* with this reference regardless of suspect-set size). $\hat{e}_f^{\text{test}}$ is the audit-split efficacy—a *diagnostic*, never an input to $m_{\text{pred}}^{\text{cal}}$. The last column is the *efficacy-transport ratio* $\exp(2|\log \hat{e}_f^{\text{cal}} - \log \hat{e}_f^{\text{test}}|)$, the symmetric factor by which the between-split efficacy shift would move the external-null effective size. $^\dagger M_{\text{eff}}^{\text{test}}$ is a plug-in inversion of the audit-split efficacy, *not* an independently observed 0.8-power sample size from permutation curves (which we have not measured here). The median efficacy-transport ratio is $2.23\times$ and 3/5 channels exceed the pre-specified $2\times$ tolerance, so the pooled calibration-to-audit transport claim is *withdrawn* (§6.2); under the finite clean reference the attainable- m gap is generally larger still (mean NLL 96 vs. $\approx 1,283$; all M_{eff} values in this paper use the z -sum $z_{.95} + z_{.80} = 2.486$ —higher-precision z values move this figure to $\approx 1,289$, immaterially).

$e_f \backslash m$	100	500	1,000	5,000
0.10	> 1	> 1	.786	.352
0.25	.995	.445	.315	.141
0.50	.497	.222	.157	.070
1.00	.249	.111	.079	.035

Table 11: Approximate smallest detectable α at size .05 and 80% power with an externally calibrated null. Entries above one mean that even full exposure is underpowered under the local approximation.

Channel	onset slope b	matches $-1/2$?
Checkpoint mixture (Eval. A)	−0.569	yes
Exact injection (Eval. B)	−0.529	yes
Surface perturbation (Eval. B)	n/a (subthreshold)	—
Answer-only (Eval. B)	n/a (subthreshold)	—

Table 12: Secondary scaling diagnostic: b is the fitted slope of $\log \hat{\alpha}_*(m)$ on $\log m$, predicted $-1/2$. The strong channels (Eval. A checkpoint mixture, Eval. B exact injection) recover slopes near $-1/2$; the surface and answer-only injections (the two subthreshold rows) are too weak to reach 0.8 power at the tested m , so no onset is defined—reported straight rather than extrapolated. Values from the audit and injection summaries.

Setting	$\hat{e}_f$	$\underline{e}_f$ (var.-adapt.)	Cert. coverage
Exact injection	0.741	0.000	0.951
Surface perturbation	0.343	0.000	0.938
Answer-only	0.150	0.000	0.953

Table 13: Point efficacy (mean over three seeds) versus the variance-adaptive 95% floor $\underline{e}_f$ on the Evaluation B injections ($n=200$ exposed/held-out per mechanism). The floor is 0 at this sample size: with only 200 items the simultaneous Hoeffding slack exceeds the mean gap, so the *certified* lower bound on ρ is vacuous even where the point efficacy is clearly nonzero—an honest small-sample limitation of the distribution-free floor, not of the point estimate. Larger calibration sets are required for a nonvacuous floor.

Field	Required report
Base checkpoint	EleutherAI/pythia-160m (Pile-trained)
Optimizer	AdamW, learning rate 5×10^{-5} , fp32, no clipping
Budget	400 continued-pretraining steps on a Pile background corpus
Contamination	exact / paraphrase / surface (word-order shuffle) / answer-only; 200 injected items ($k=4$ copies each) for exact, surface, and answer-only, but only the 138 validation-accepted paraphrase items for the paraphrase arm—so mechanism arms are <i>not</i> dose-matched in items or contamination tokens, and the ordering is descriptive; SQuAD source
Matched control	clean $\alpha=0$ run, same steps and background data
Stopping	Fixed before evaluation; no audit-driven checkpoint selection
Seeds	training seeds 0, 1, 2 (initial) and 0–4 paired with clean continuations (v2); all reported, no runs discarded; 500 Monte-Carlo replicates (Eval. A), 150–200 trials per seed (Eval. B)

Table 14: Minimum training-run disclosure.

Method	Primary deliverable	Prospective?	Access
Loss / Min-K% / neighborhood	item-level membership score	no	gray-box
Dataset Inference (Maini et al., 2024)	dataset-level p -value	no	gray-box
PaCoST (Zhang et al., 2024)	paired-confidence p -value	no	black-box
Proving contamination (Oren et al., 2024)	exchangeability p -value	no	black-box
ConStat (Dekoninck et al., 2024b)	performance inflation	no	black-box + refs
FTD (Zhang et al., 2026)	FDR-controlled decision set	no	gray-box
DCQ (Golchin and Surdeanu, 2025)	contamination estimate	no	black-box
This paper (audit)	power at (α, m) ; budget m_{plan}	yes	matched controls
This paper (certificate)	lower bound $\underline{\alpha}$ on exposed fraction	no	matched controls

Table 15: What each line of work returns. “Prospective” means the quantity is computable *before* the audit split is opened and predicts what the audit will be able to detect. All methods listed answer legitimate and different questions; the column that is empty above the rule is the one a non-rejection needs in order to be interpretable. Our prospective column carries the caveat established in §6.2: the efficacy-only Gaussian budget is miscalibrated at small m , and the deliverable is the simulated planner’s budget, under a guarantee conditional on the empirical calibration distribution.